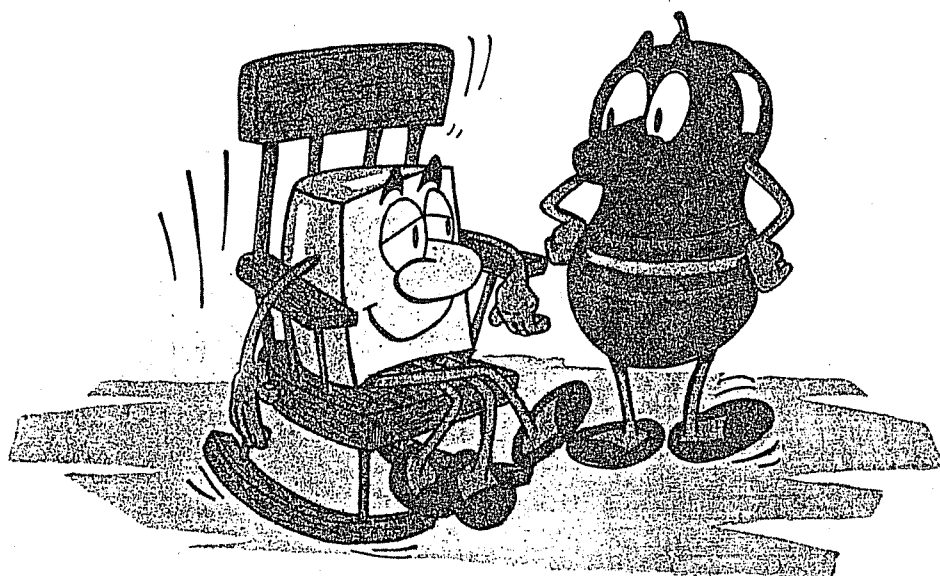


Affalé sur le rocking-chair, c'est TRANSI; debout à côté de lui, c'est RESI. De gags en coups fumeux, ils font d'une matière réputée rébarbative, la plus désopilante débandade d'électrons. Tout au long de cet album, nos deux facétieux composants électroniques vous entraînent à la découverte de leur univers mystérieux. Ils sont parfois à "contre-courant", ce qui ne va pas sans créer quelques "tensions", pour votre plus grand plaisir.



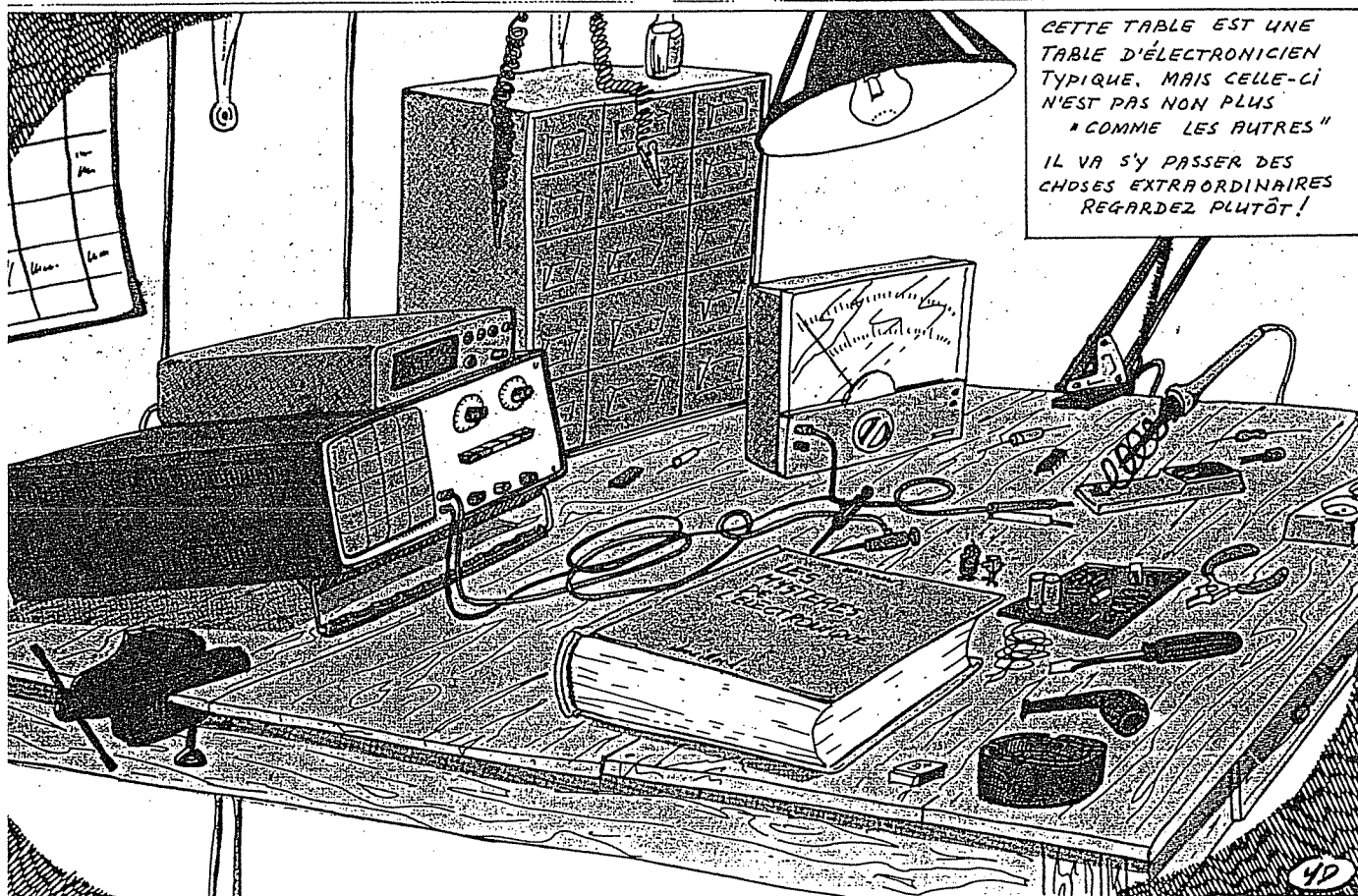
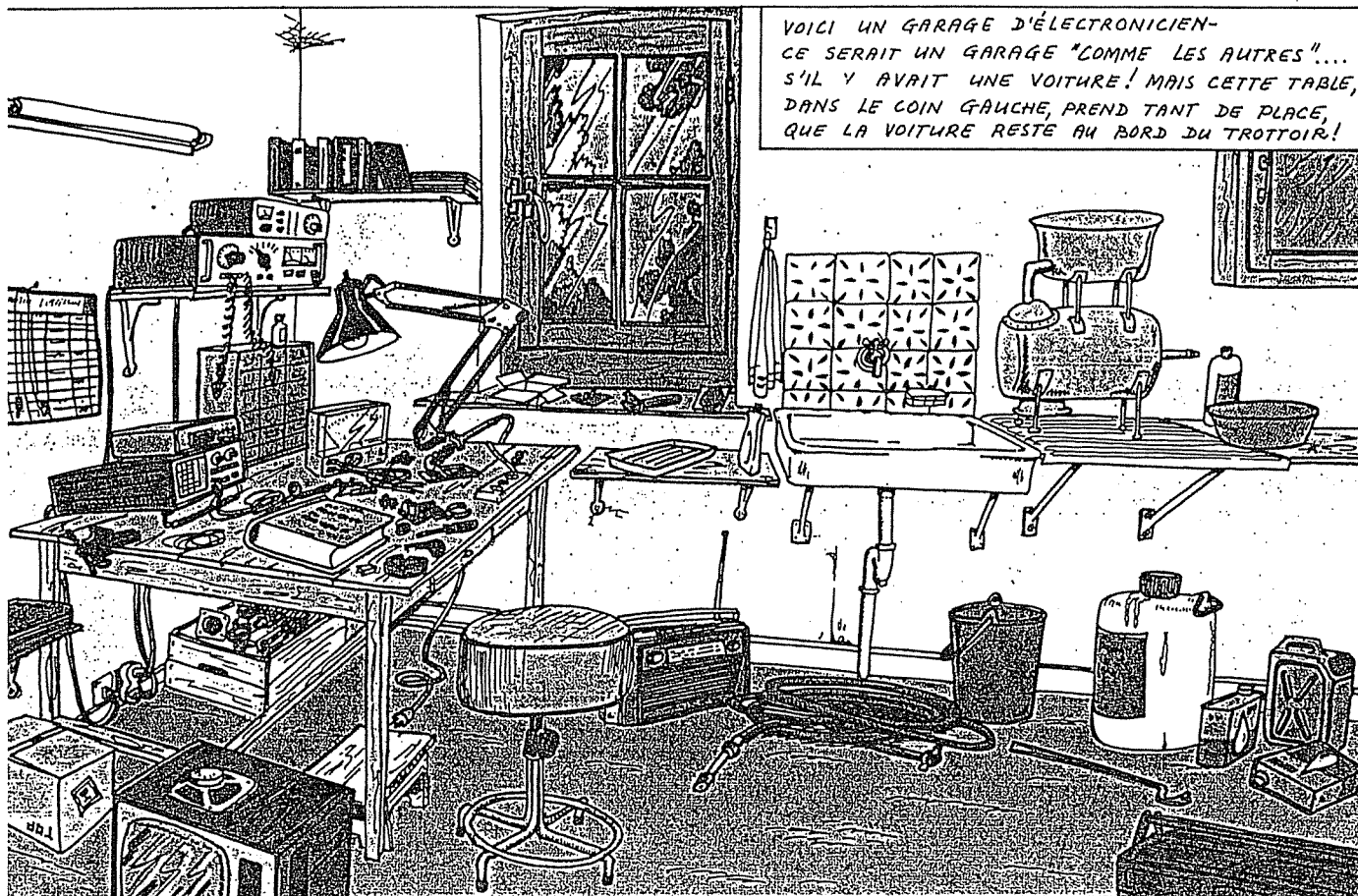
D'autres aventures les attendent, où retentiront les rires pincés de quelques diodes, les roucoulements d'une LED cabotine, et les moqueries explosives de certains condensateurs . . .

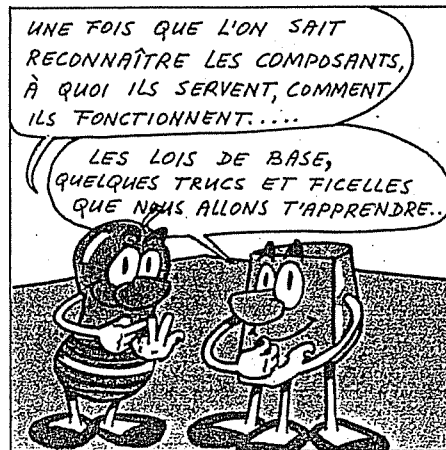
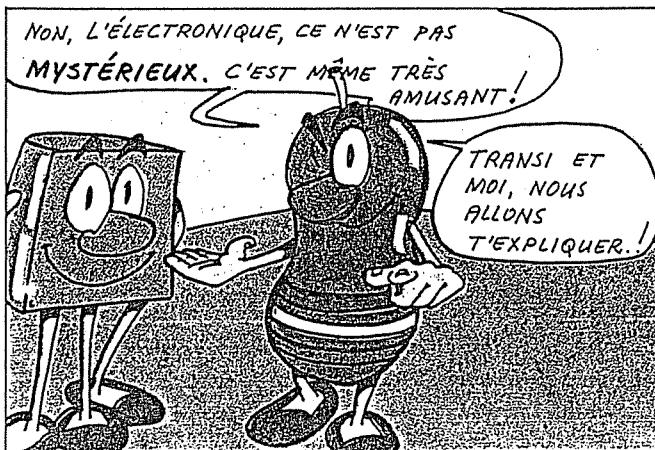
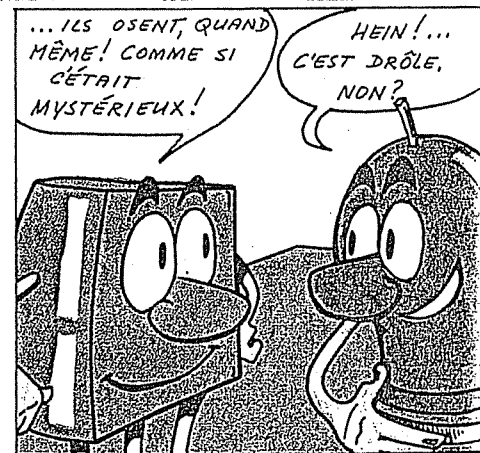
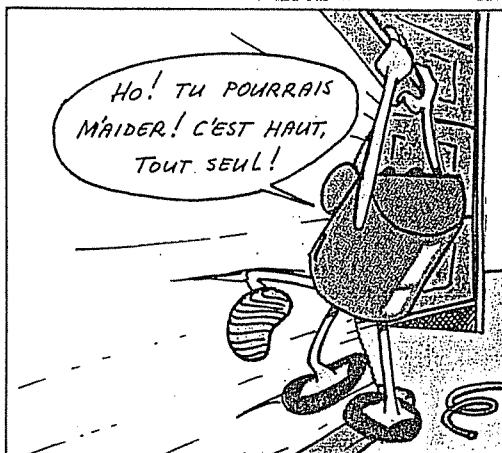
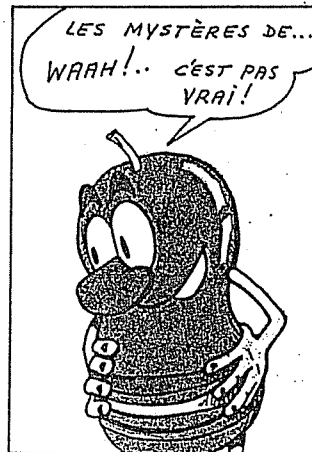
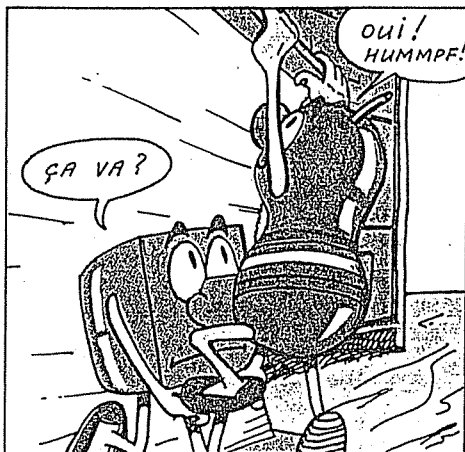
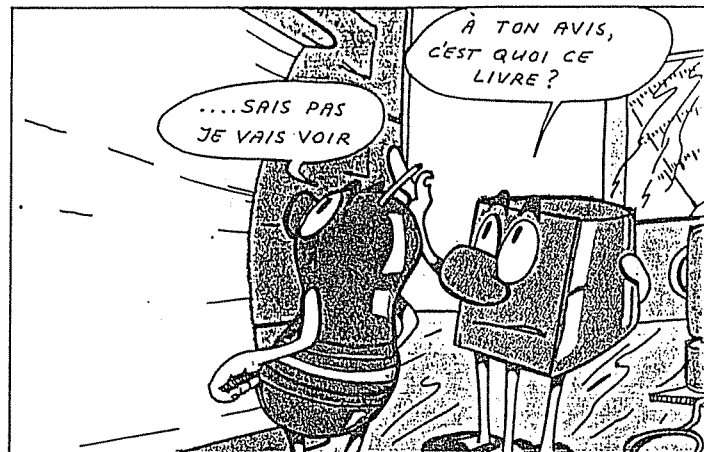
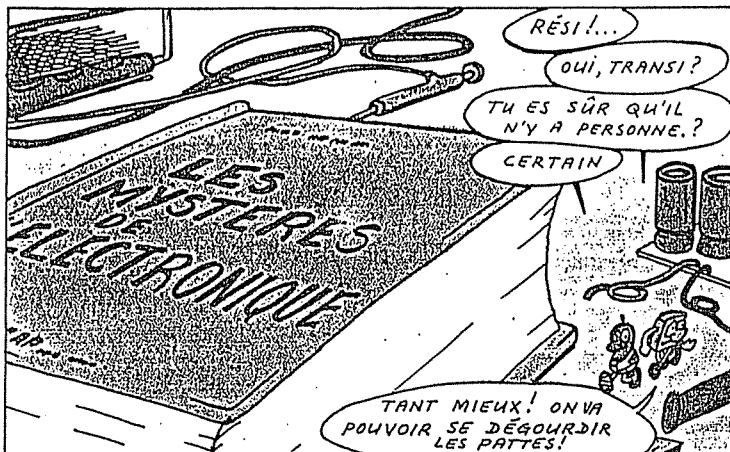
Suivez les aventures de RESI et TRANSI chaque mois dans la revue ELEX - Explorez l'Électronique !

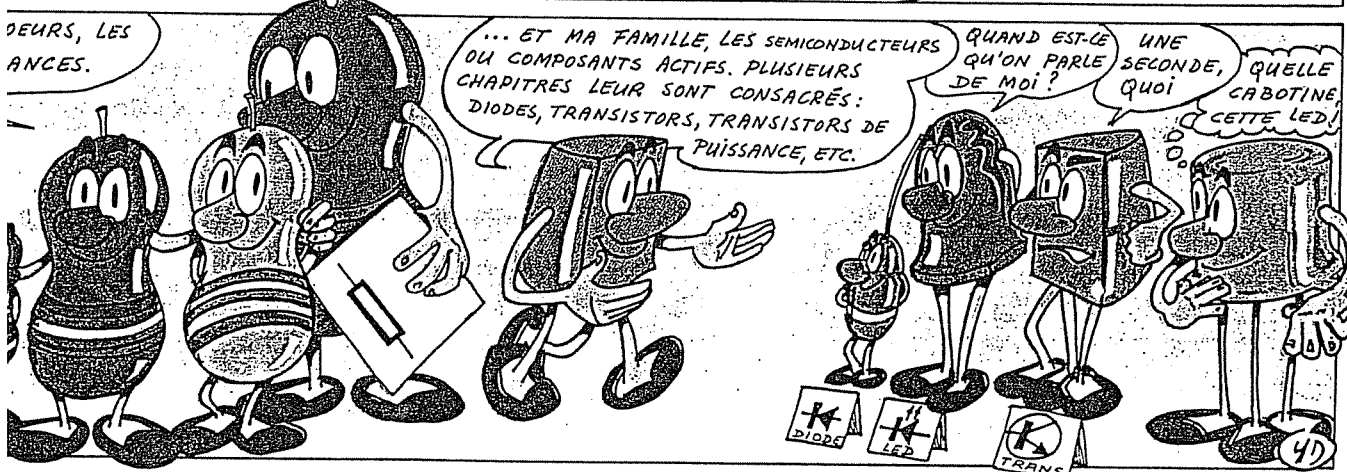
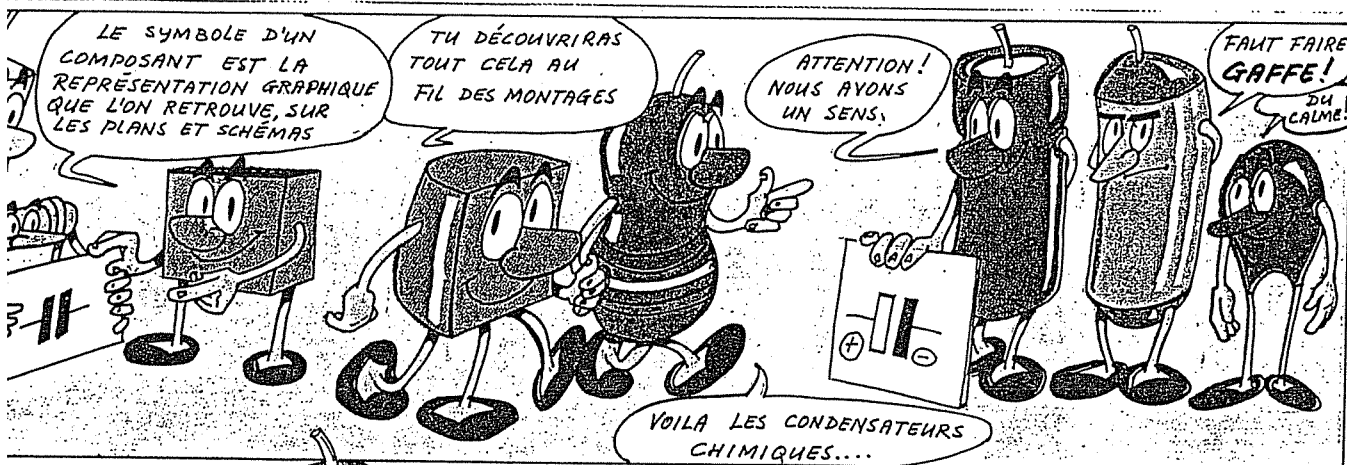
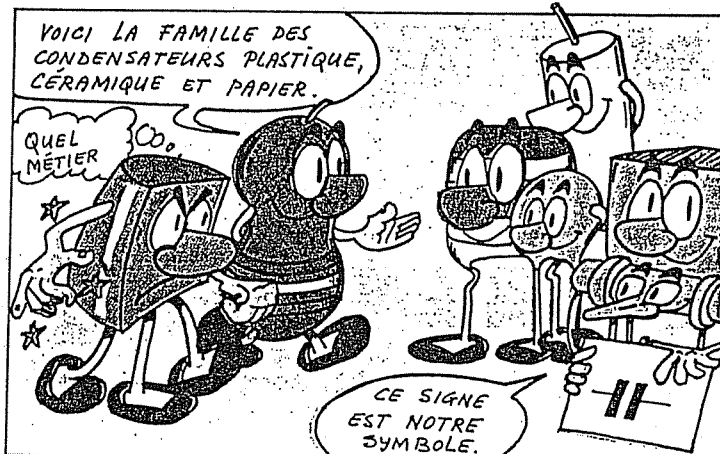
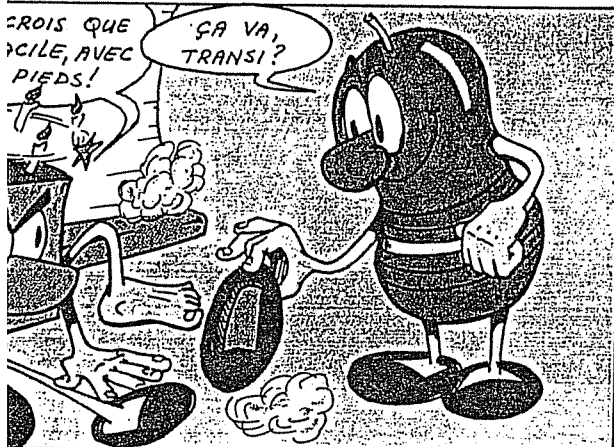
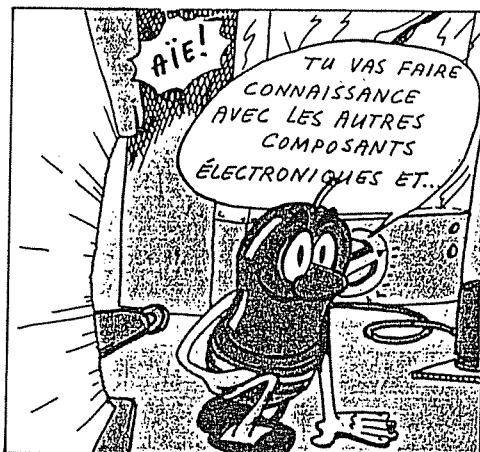
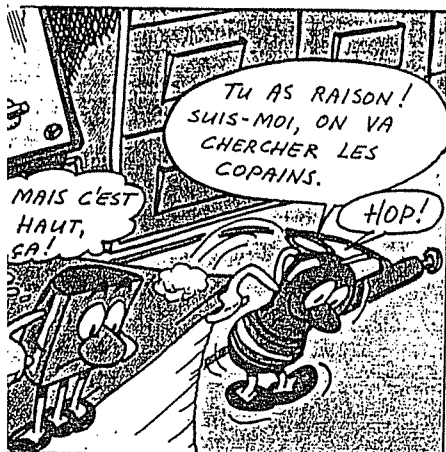
RESI stor & **TRANSI**[®] stor:

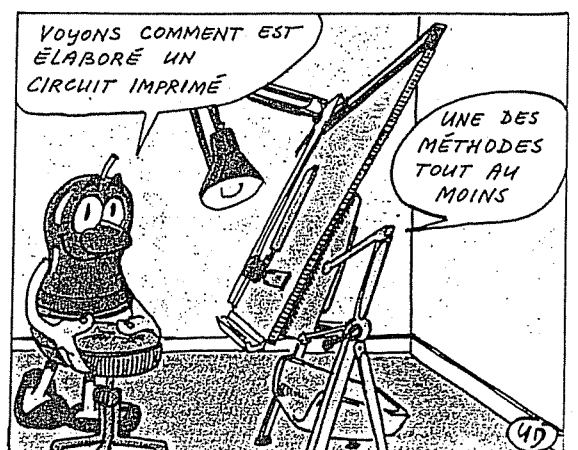
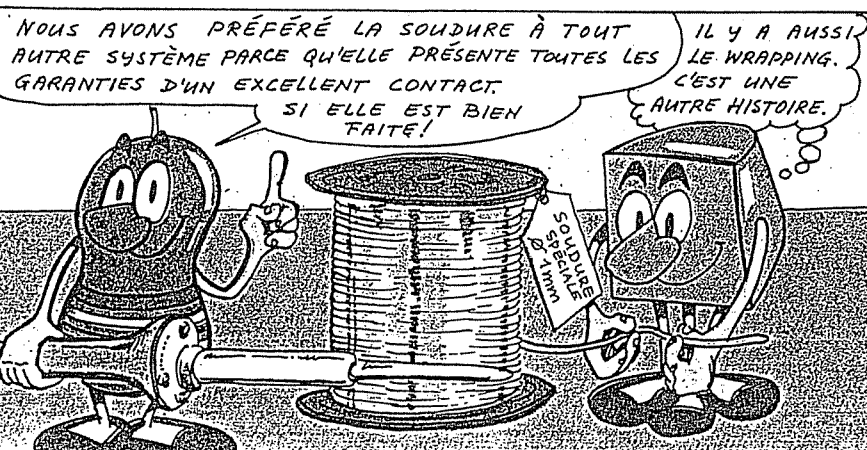
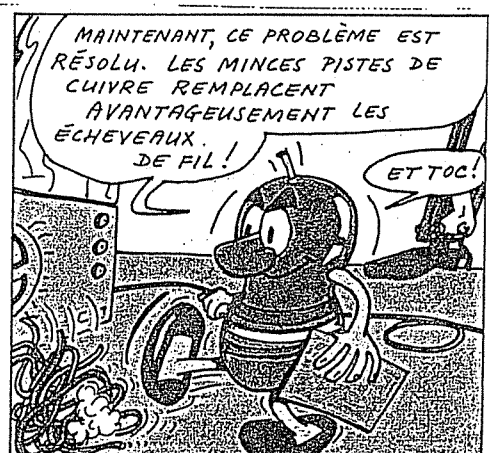
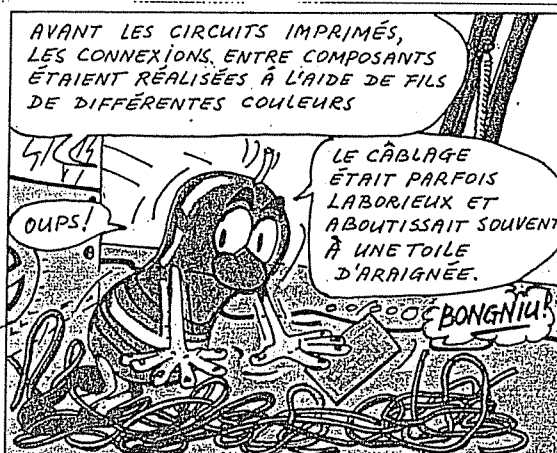
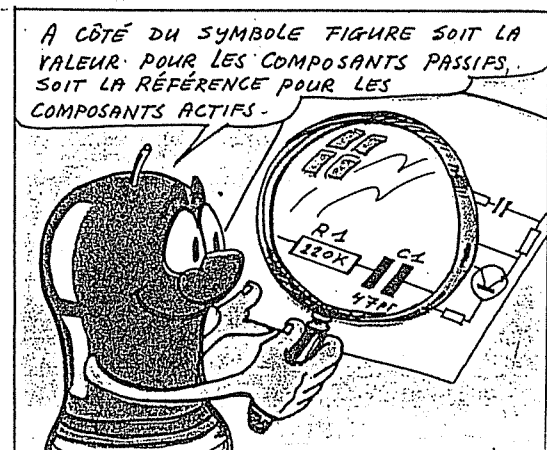
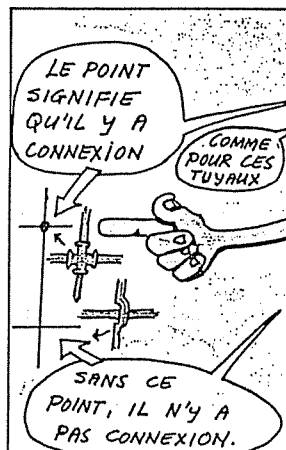
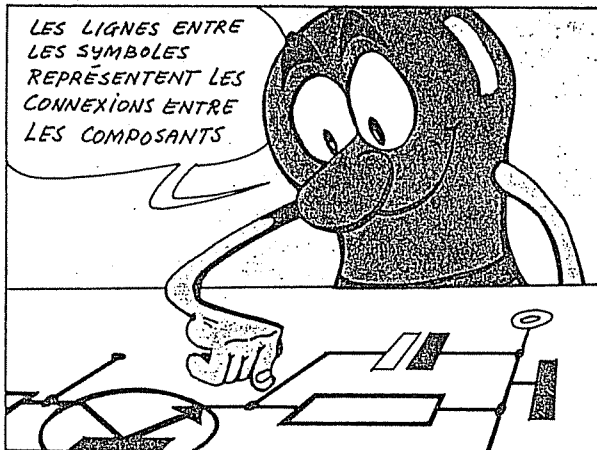
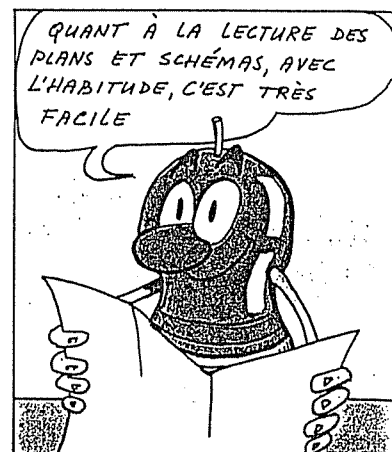
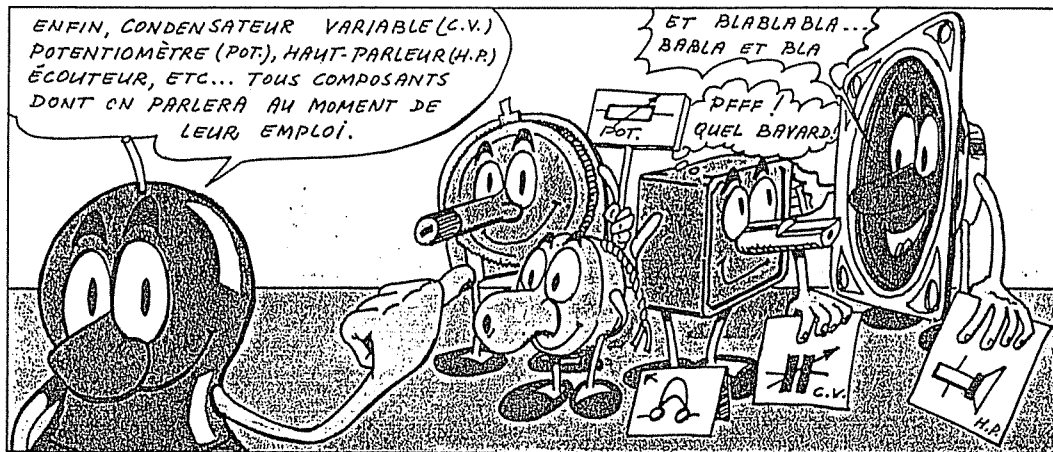
RESI et TRANSI font échec aux mystères de l'électronique avec un
TESTEUR DE CONTINUITE, UN MANIPULATEUR DE MORSE et
UN AMPLIFICATEUR, A CONSTRUIRE SOI-MEME!

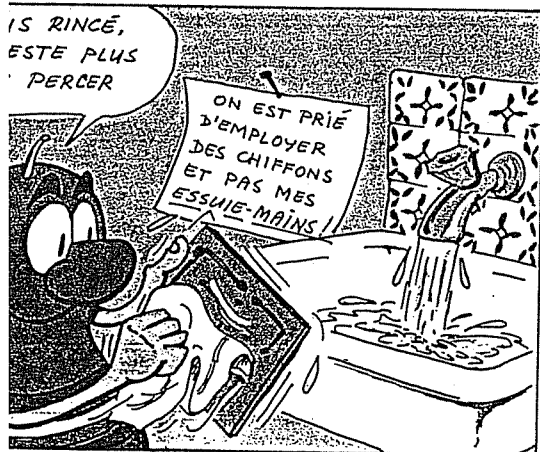
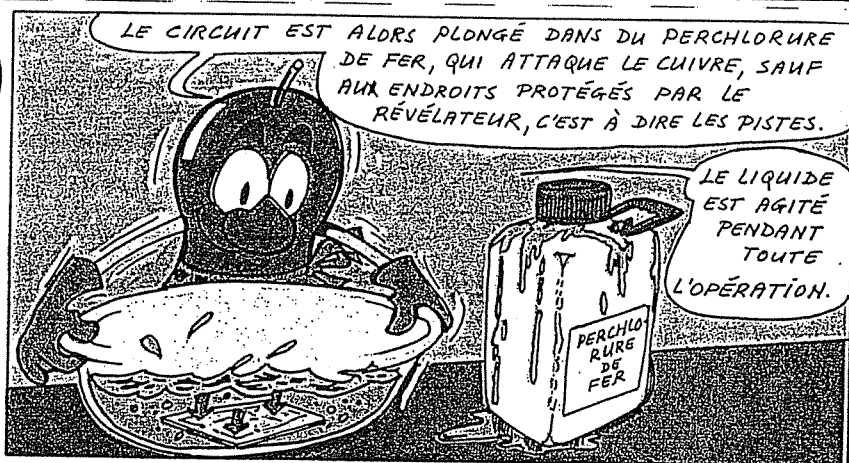
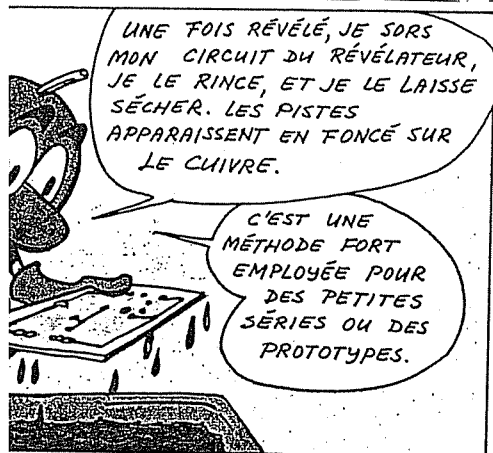
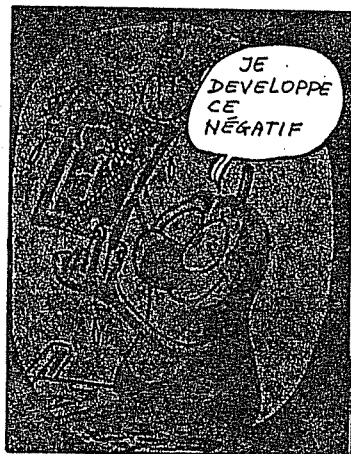
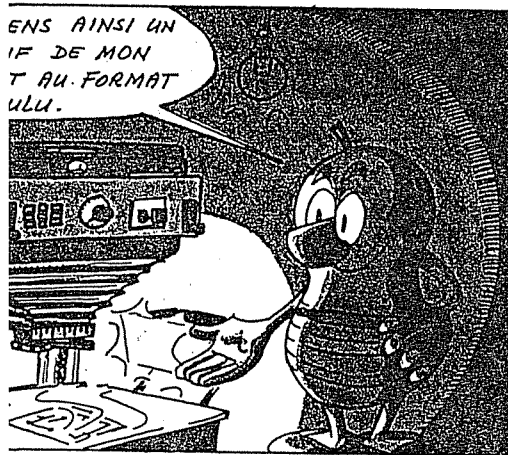
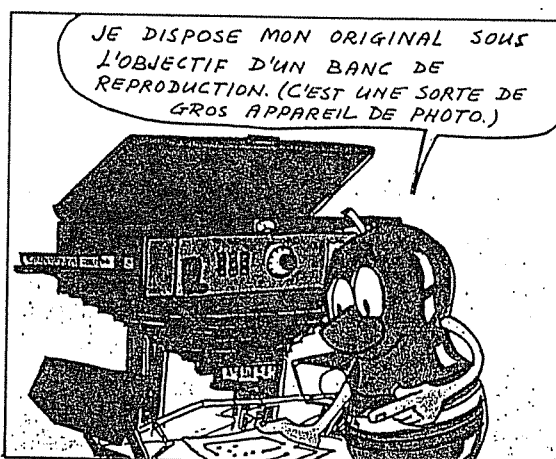
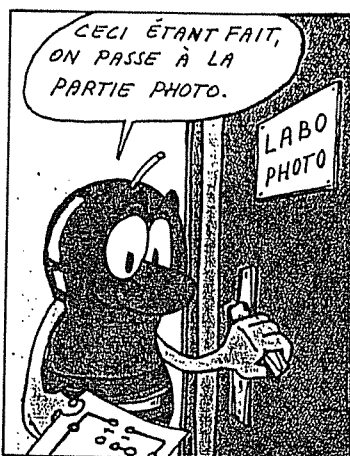
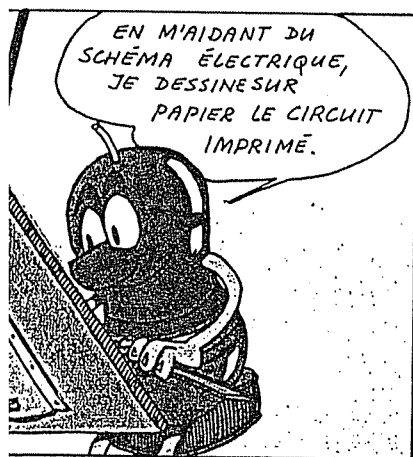
PUBLITRONIC

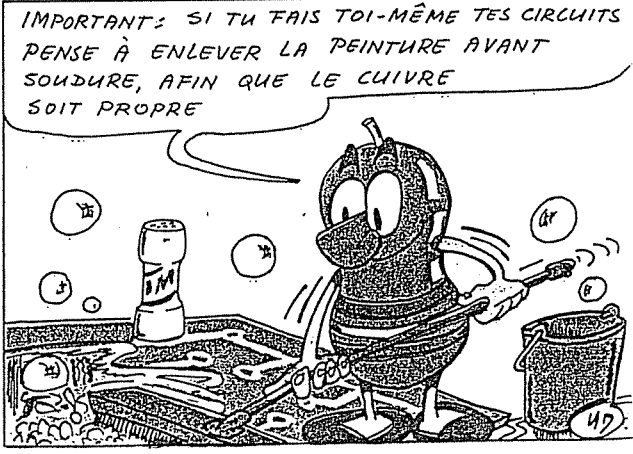
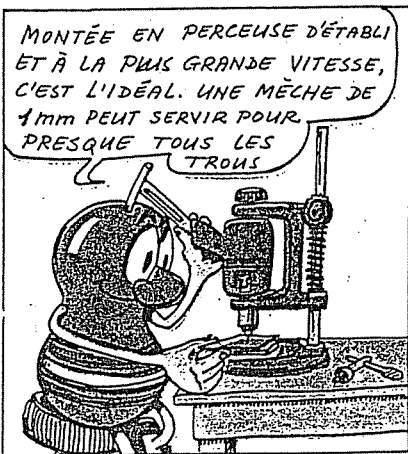
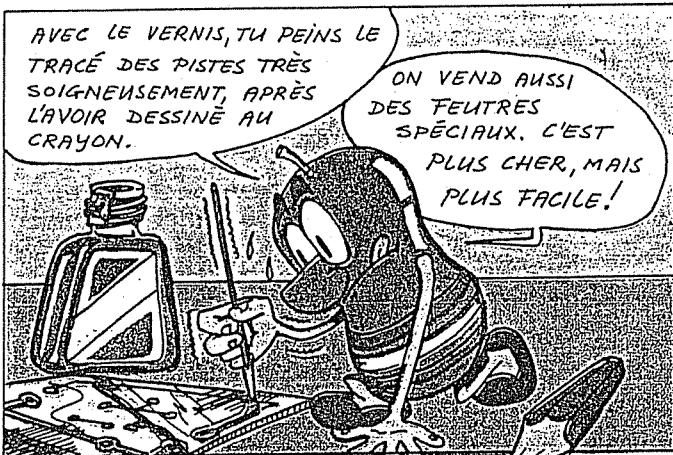
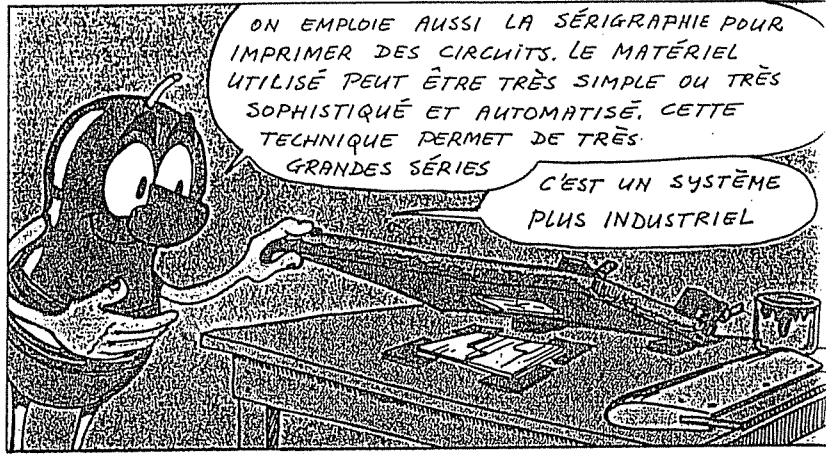
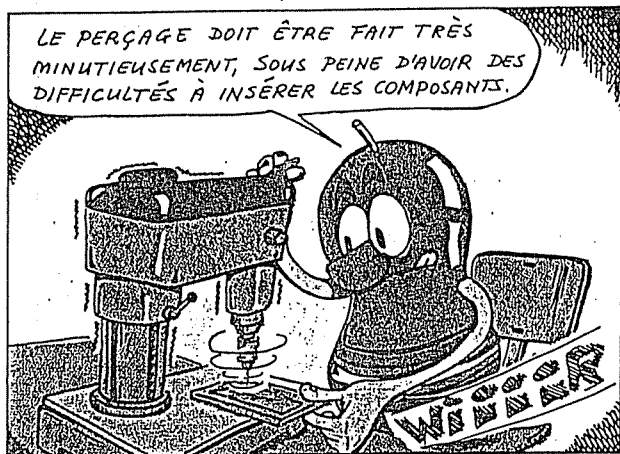




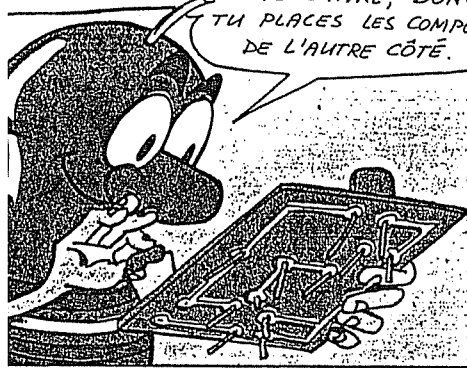




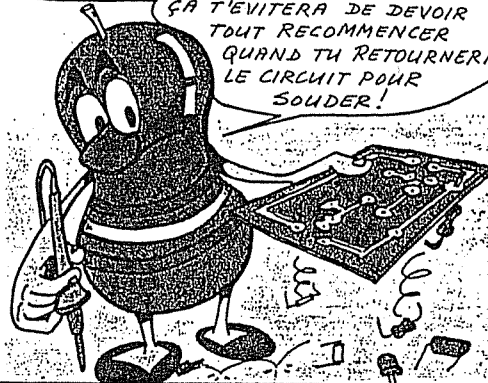




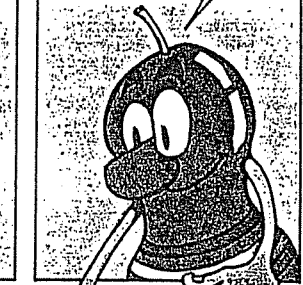
COMPOSANTS SE PLACENT SUIVANT LES
MAS D'IMPLANTATION. ATTENTION: LES
URES SE FONT COTÉ CUIVRE, DONC
TU PLACES LES COMPOSANTS
DE L'AUTRE CÔTÉ.



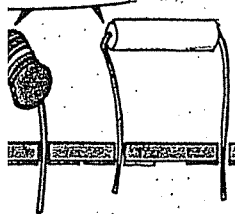
UN CONSEIL: TU CAMBRES LÉGÈRE-
MENT LES PATTES DES COMPOSANTS
ÇA T'ÉVITERA DE DEVOIR
TOUT RECOMMENCER
QUAND TU RETOURNERAS
LE CIRCUIT POUR
SOUDER!



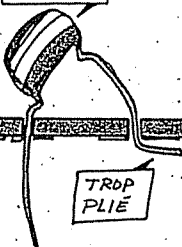
POUR BIEN FAIRE
TON CÂBLAGE,
SUIS CES EXEMPLES
DE POSITIONS DES
COMPOSANTS.



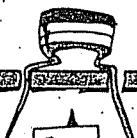
MAUVAIS



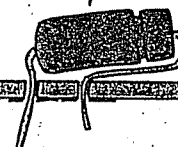
MAUVAIS



BON

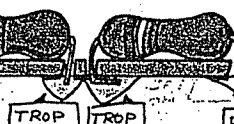


MAUVAIS



TROP LONG

QUAND À LA
LONGUEUR DES
PATTES APRÈS
SOUDURES, VOICI
D'AUTRES EXEMPLES.



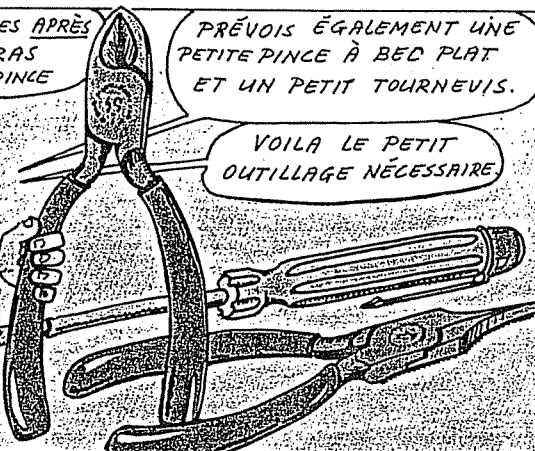
BON

DUPES LES PATTES APRÈS
SOUDÉ. TU AURAS
N D'UNE PETITE PINCE
INTE.



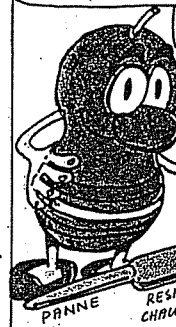
PRÉVOIS ÉGALEMENT UNE
PETITE PINCE À BÈC PLAT
ET UN PETIT TOURNEVIS.

VOILA LE PETIT
OUTILLAGE NÉCESSAIRE.

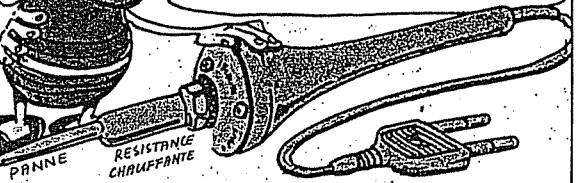


TON MARCHAND DE COMPOSANTS
TE CONSEILLERA UTILEMENT
POUR L'ACHAT D'UN BON FER
À SOUDER. C'EST L'OUTIL LE
PLUS IMPORTANT
DE TA PANOPLIE.

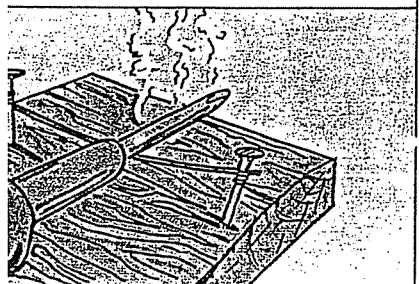
UNE PUISSANCE
DE 30 À 50W
EST
IDÉALE.



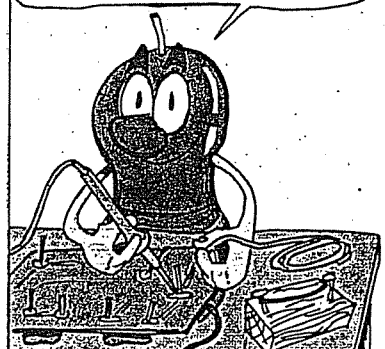
AVOIR DES SOUDURES BIEN
FAITES EST IMPORTANT: 90% DES
PANNES SONT DUES À DES "COLLAGES".



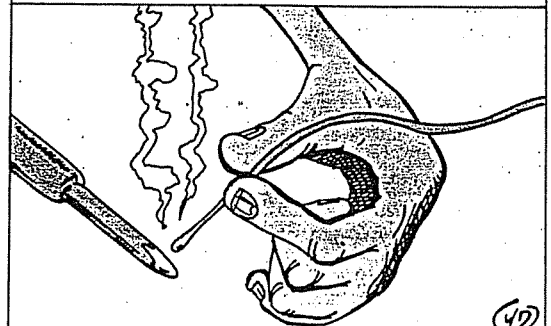
UN FER EST VENDU AVEC UN
PORT, TANT MIEUX. SINON, TU PEUX
BRICOLER UN: UNE PLANCHE,
DUS ET UN BOUT DE FIL DE FER.
DITIÉ POUR LA TABLE!

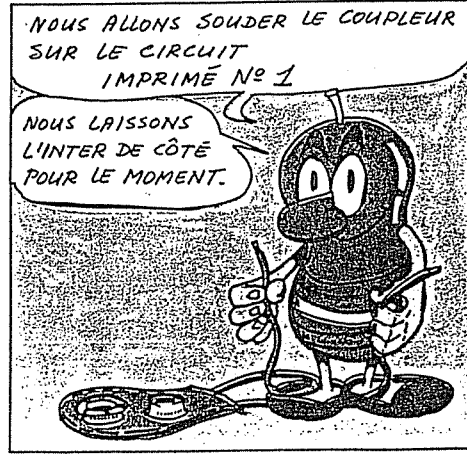
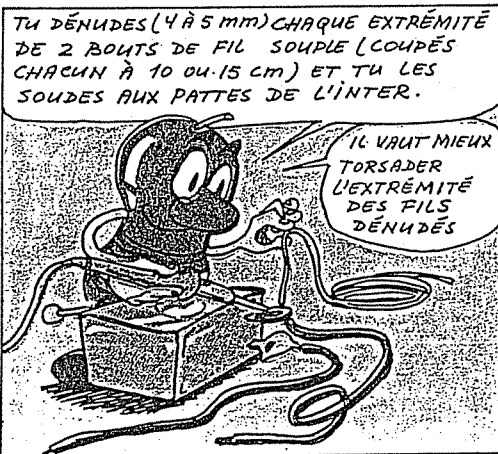
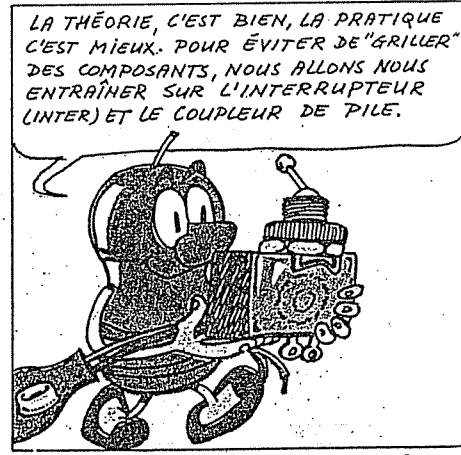
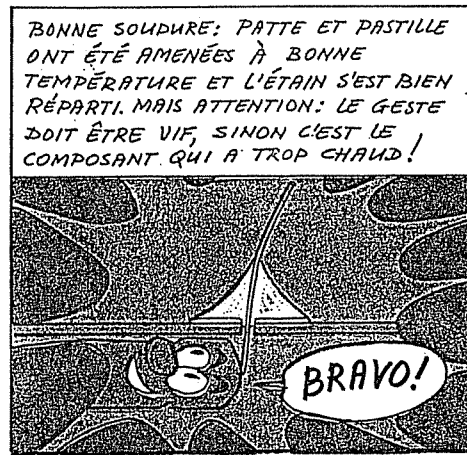
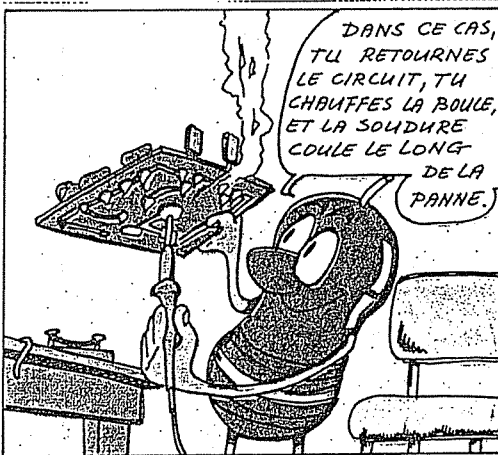
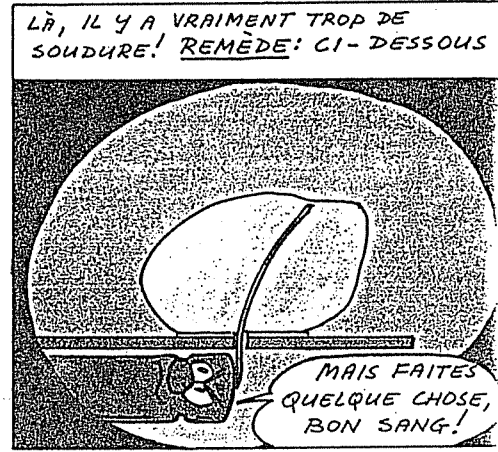
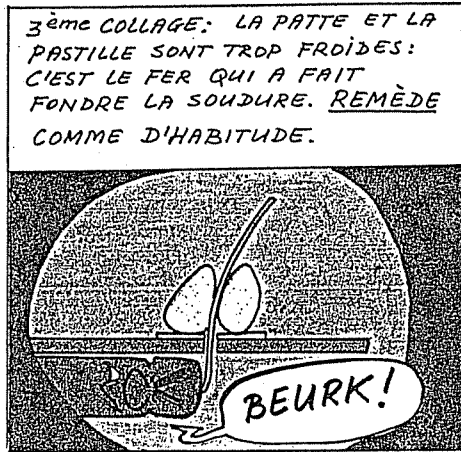
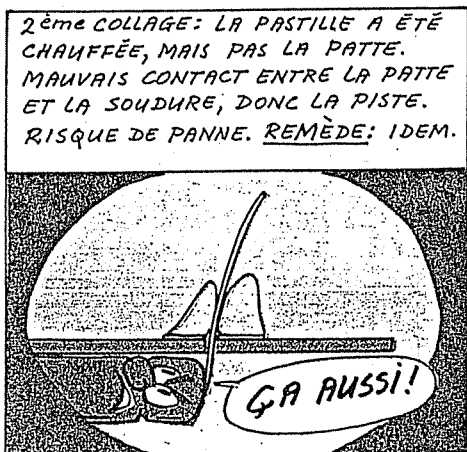
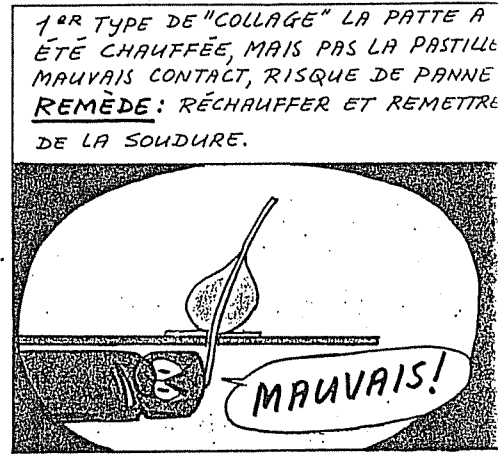
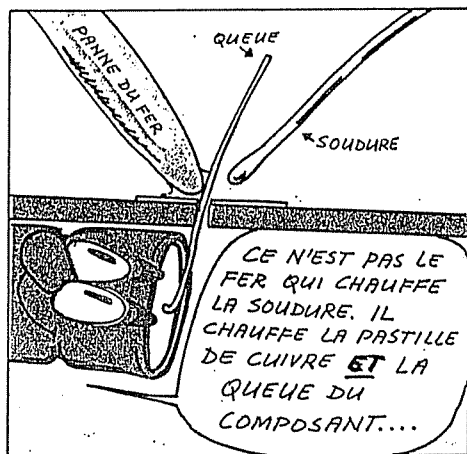


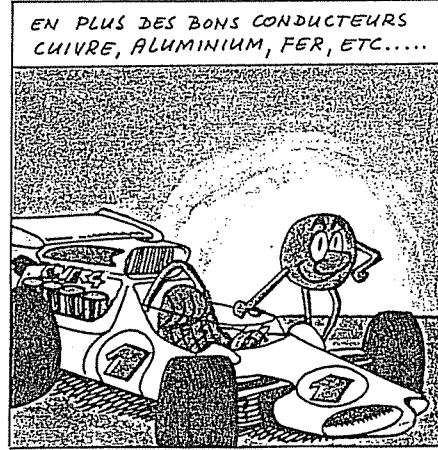
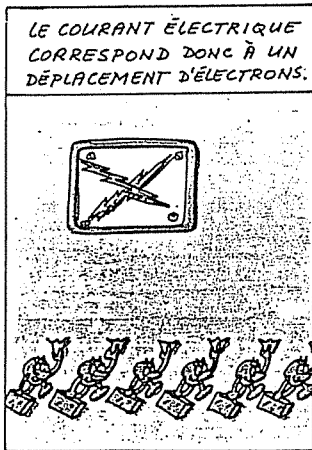
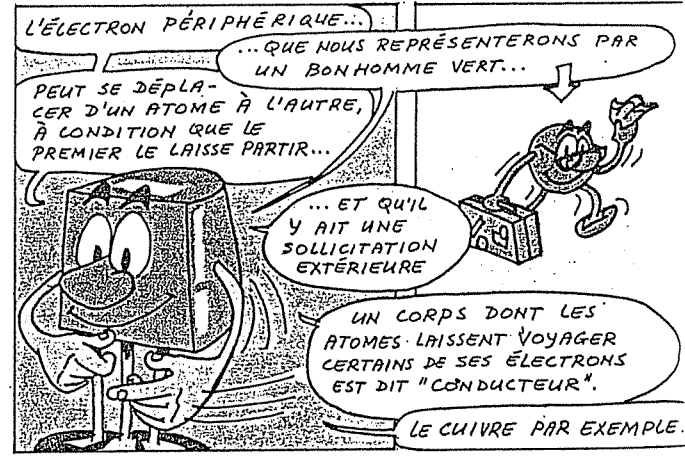
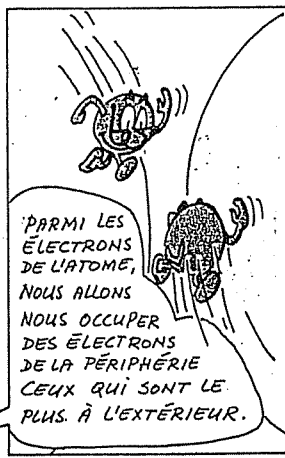
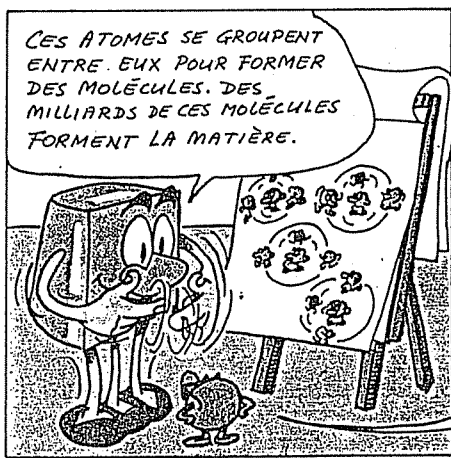
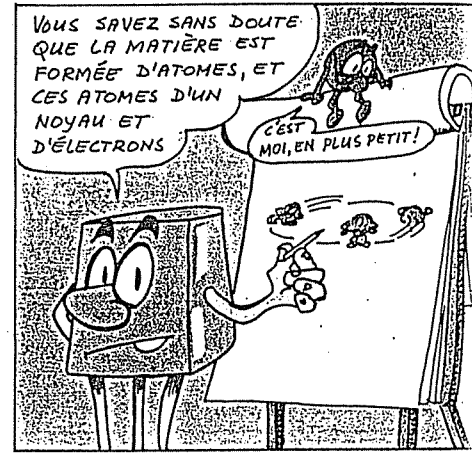
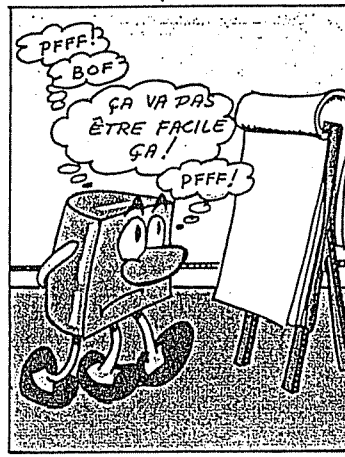
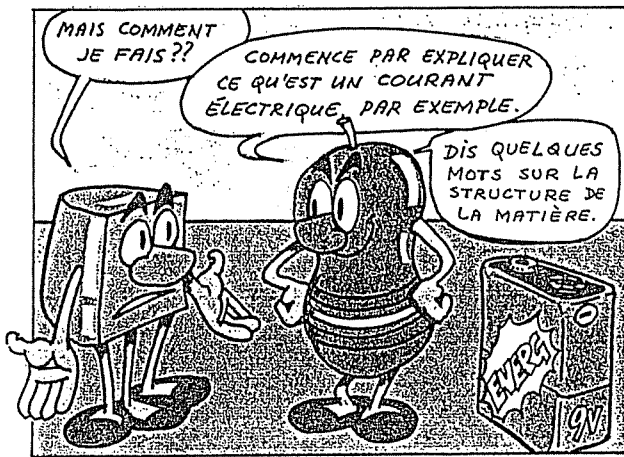
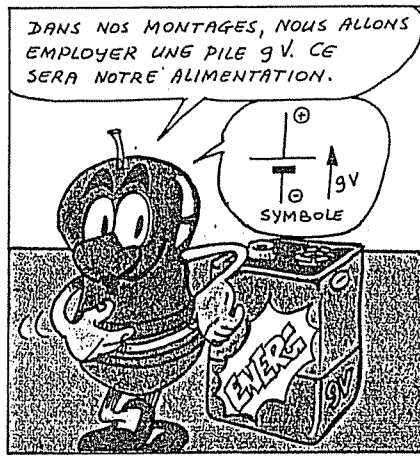
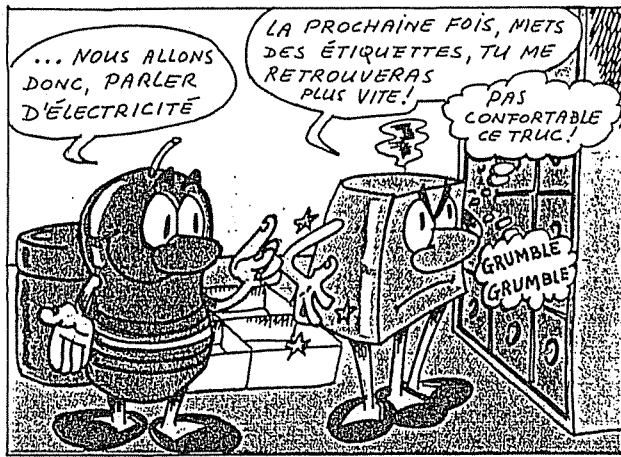
SUIS MOI BIEN: SOUDURE DANS
UNE MAIN, FER DANS L'AUTRE,
COMME UN STYLO.



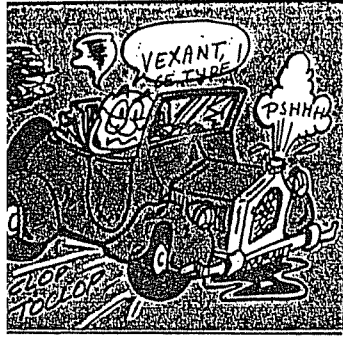
PREMIER TEMPS: TU ÉTAMES TA PANNE.
C'EST À DIRE QUE TU "MOUILLES"
LÉGÈREMENT SON EXTRÉMITÉ AVEC UNE
GOUTTE D'ÉTAIN FONDU. PAS TROP.



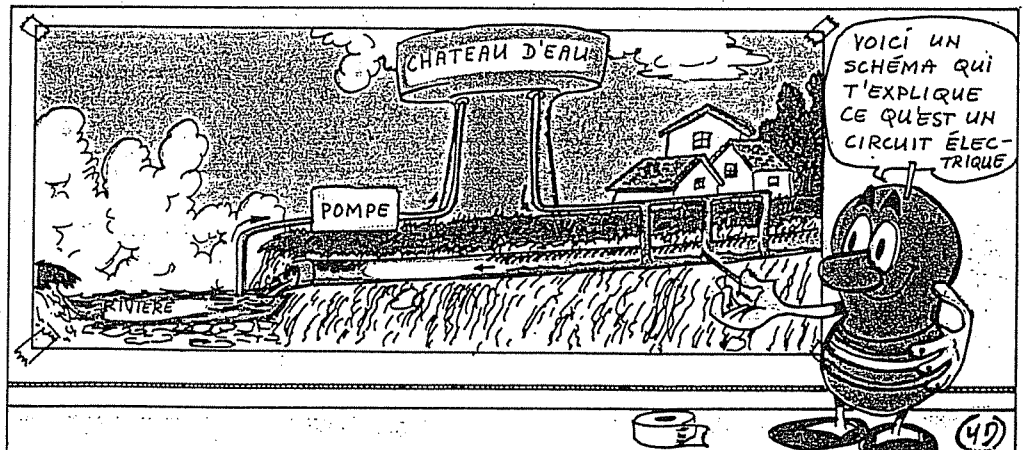
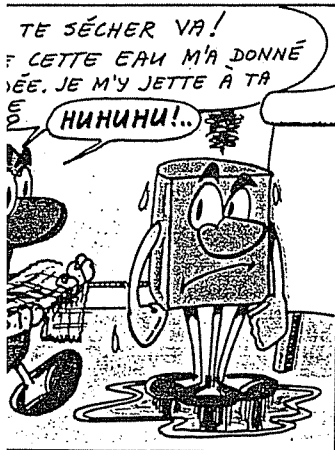
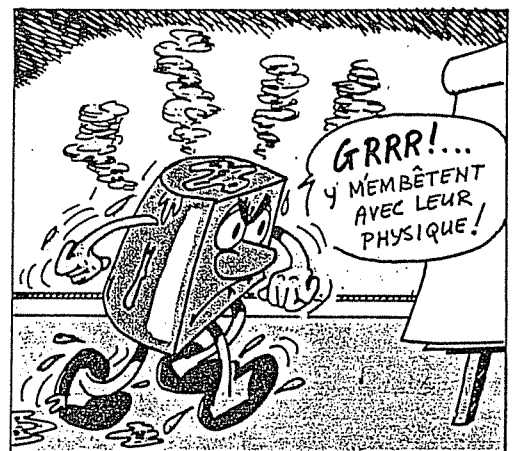
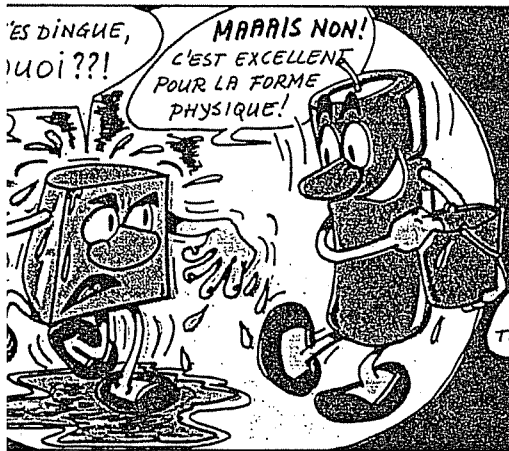
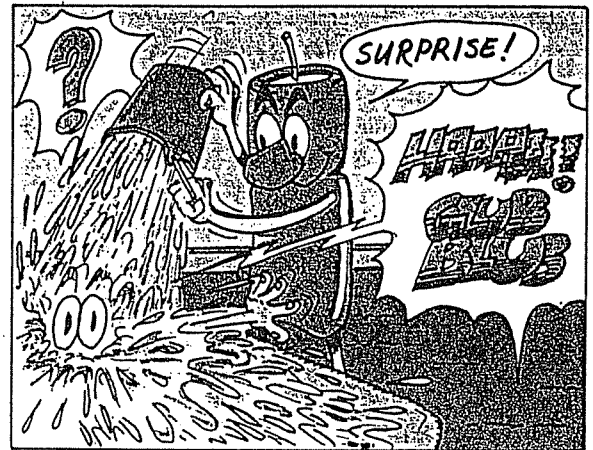
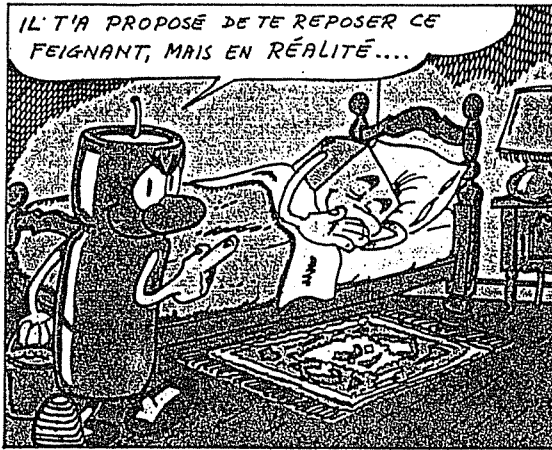
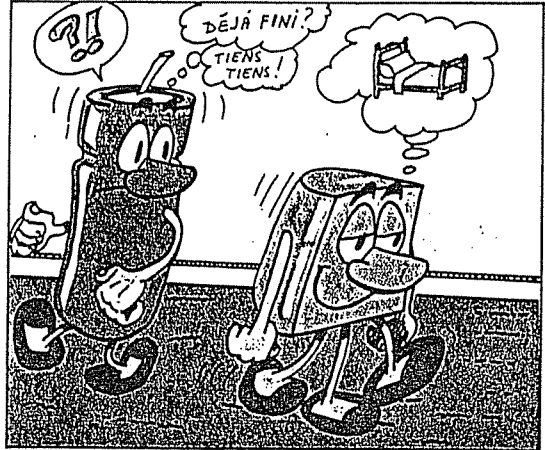
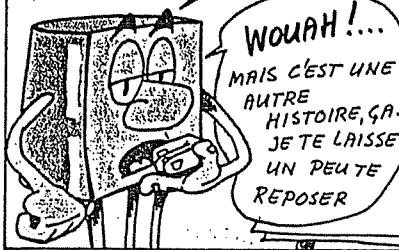


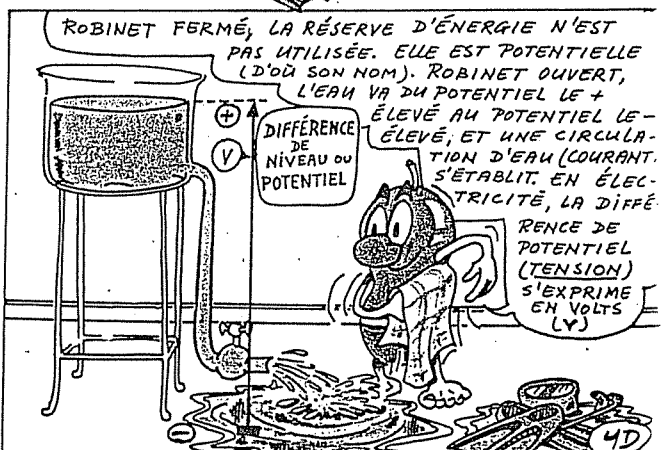
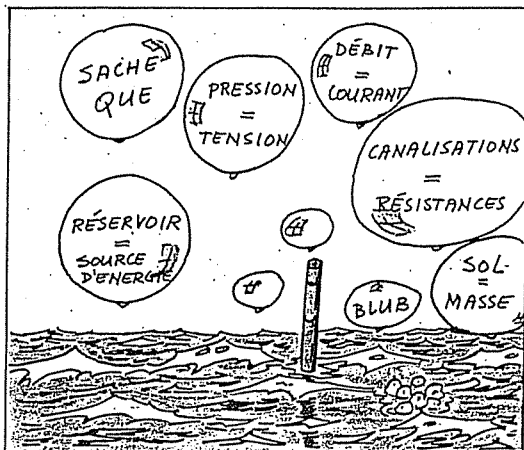
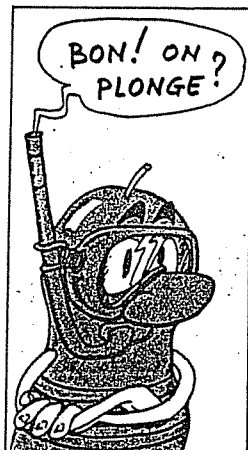
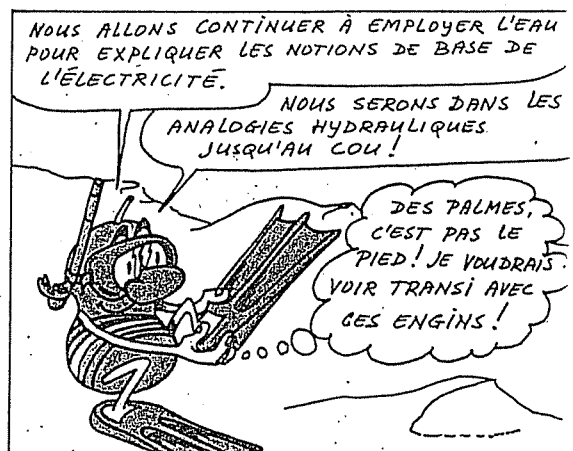
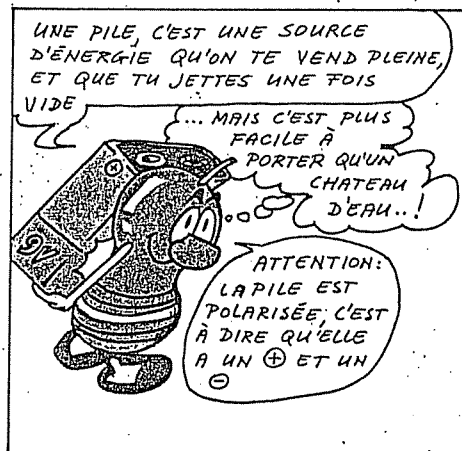
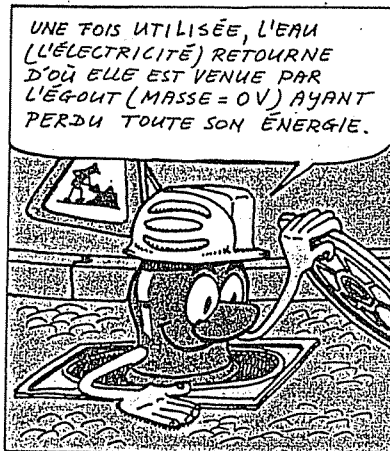
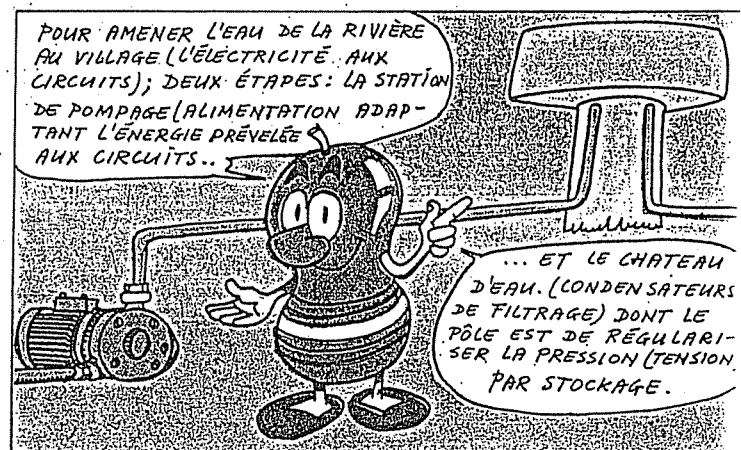
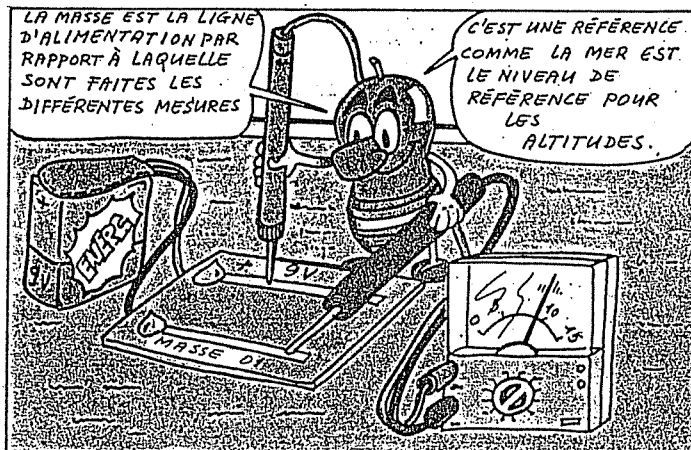
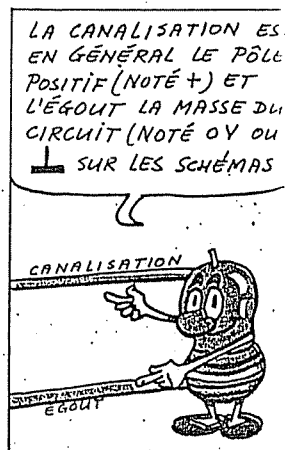
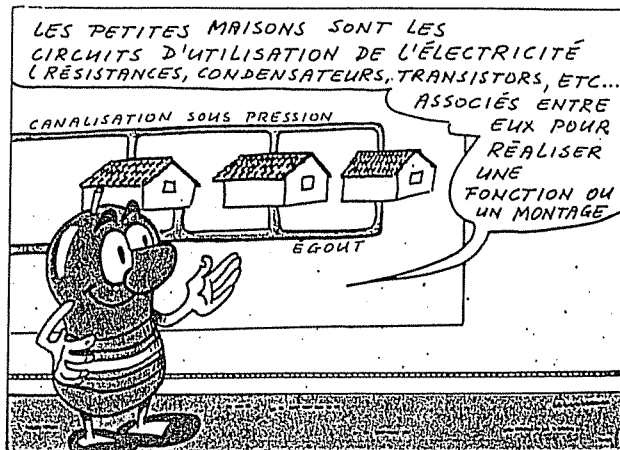
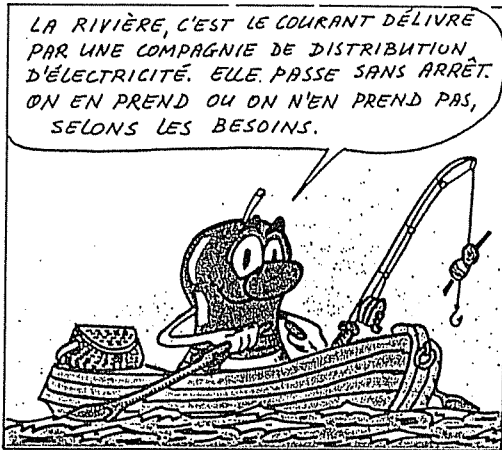


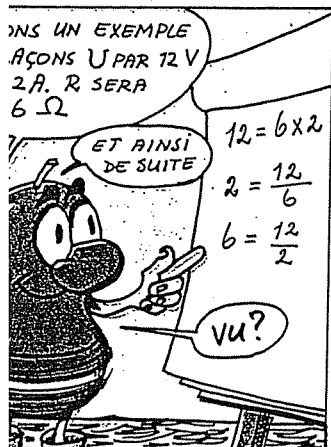
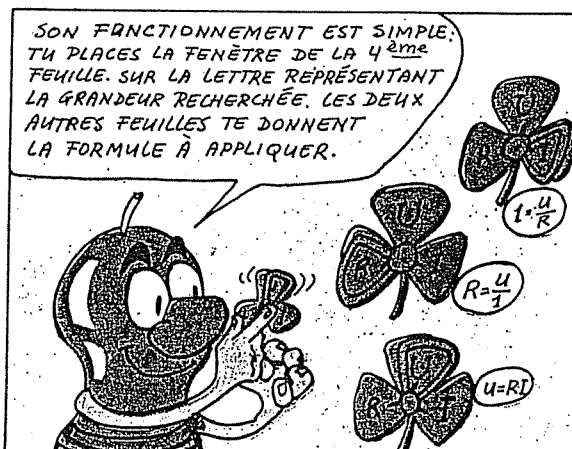
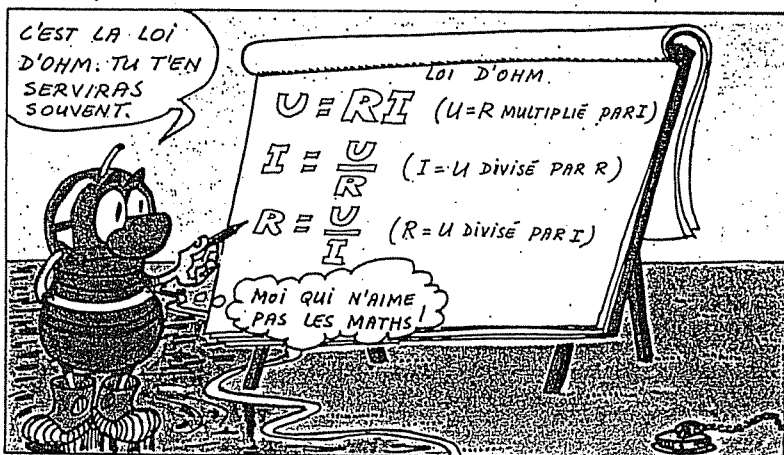
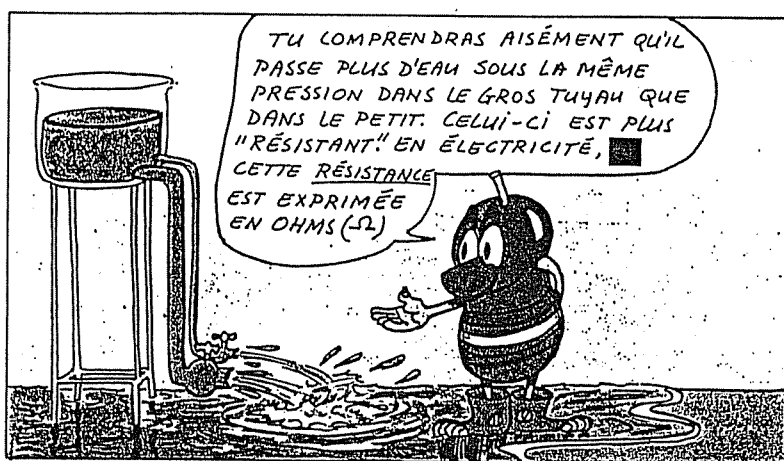
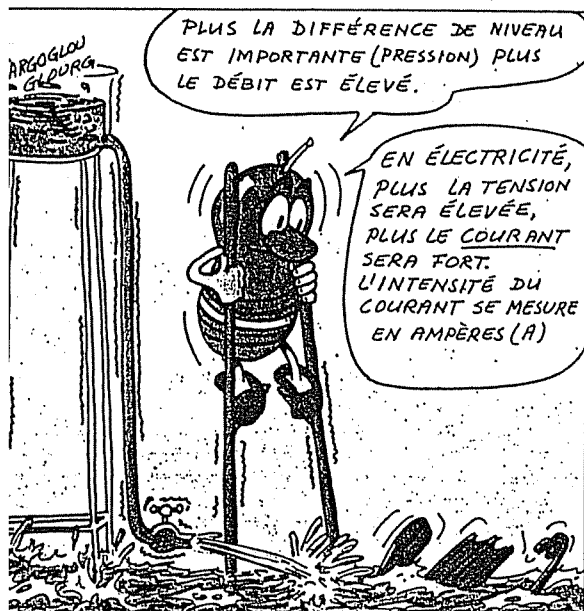
EXISTE CE QU'ON APPELLE
"SEMICONDUCTEURS"
(GALLIUM, SILICIUM....)
(LES CONDUCTEURS DU DIMANCHE).

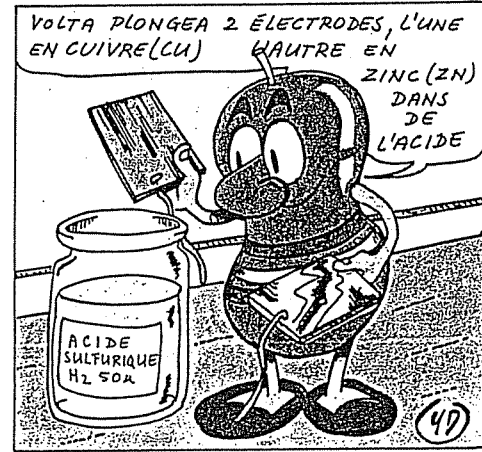
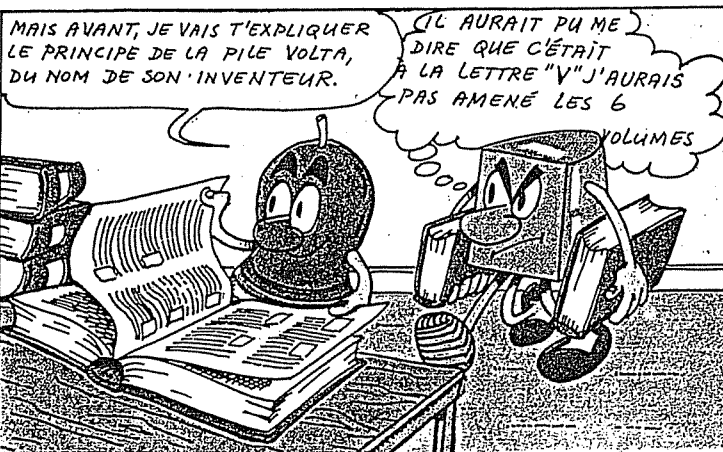
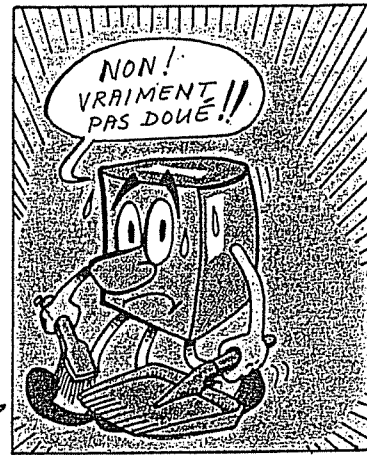
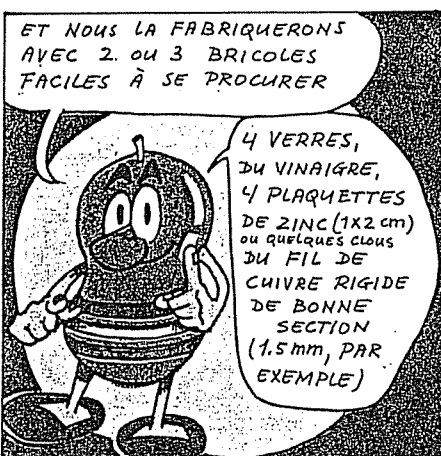
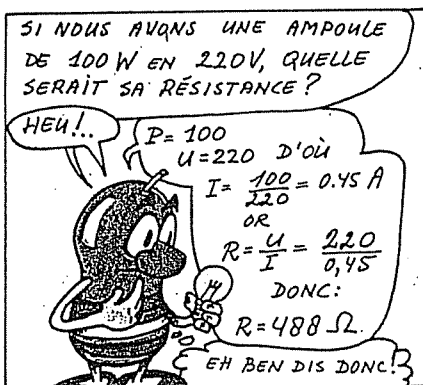
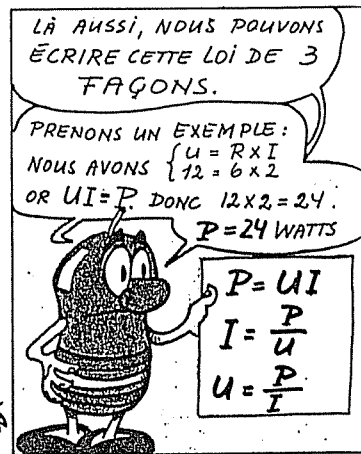
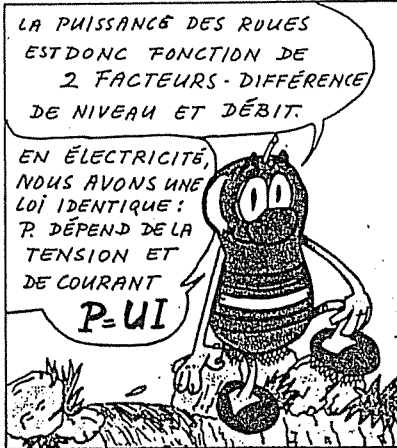
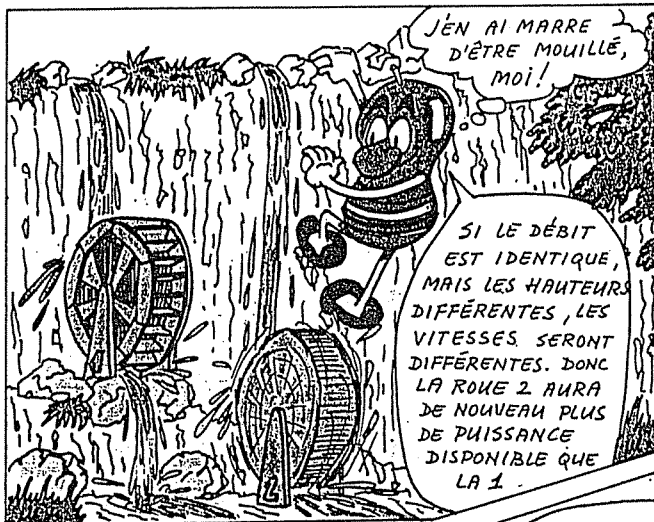


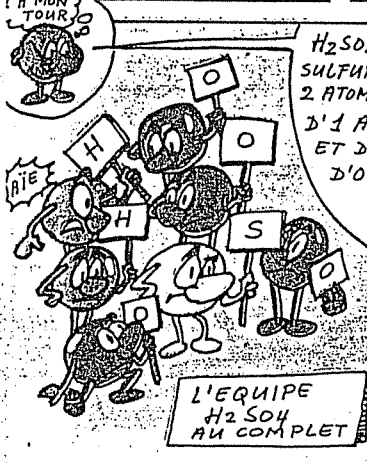
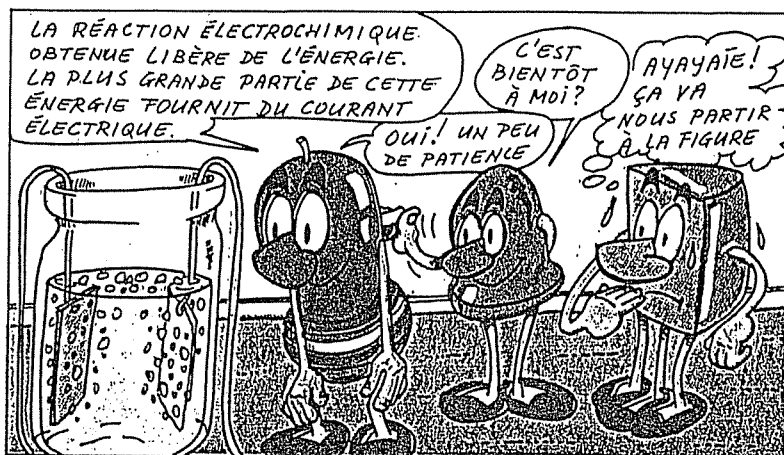
NOUS VERRONS PLUS LOIN QU'ILS
SERVENT À LA FABRICATION DES
TRANSISTORS, DIODES, DIACS ETC...
C'EST POURQUOI CES COMPOSANTS
PORTENT, PAR EXTENSION LE NOM
DE "SEMICONDUCTEURS."





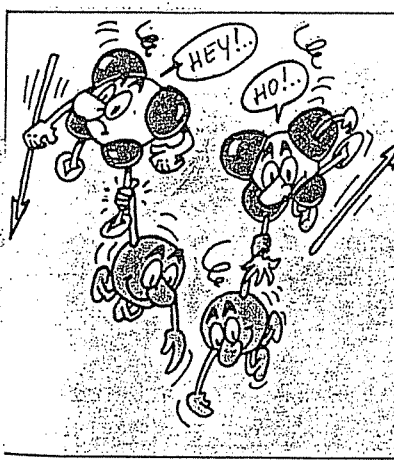




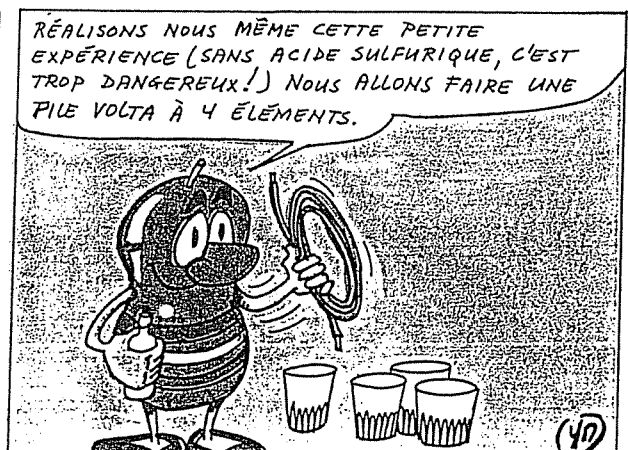
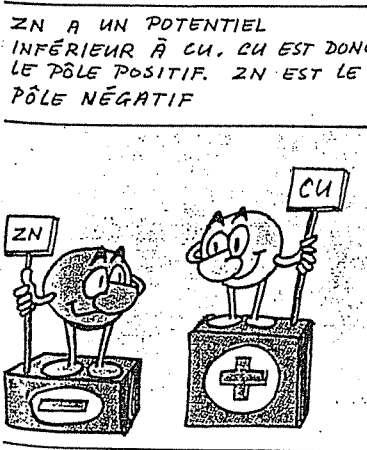
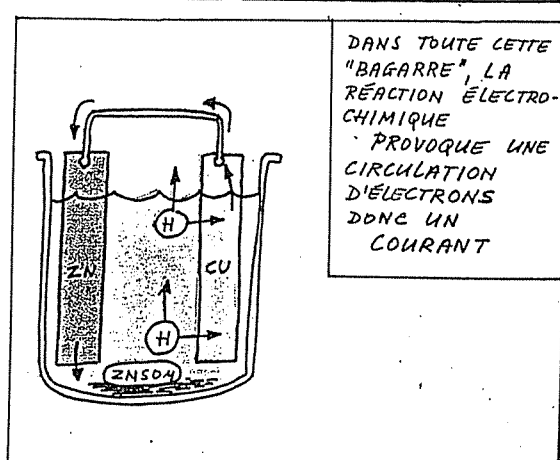


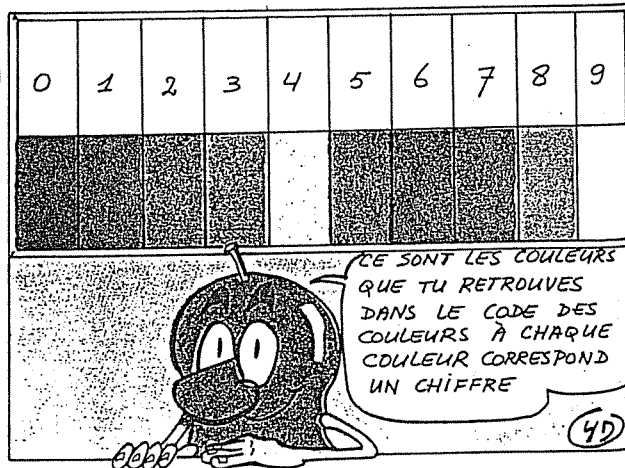
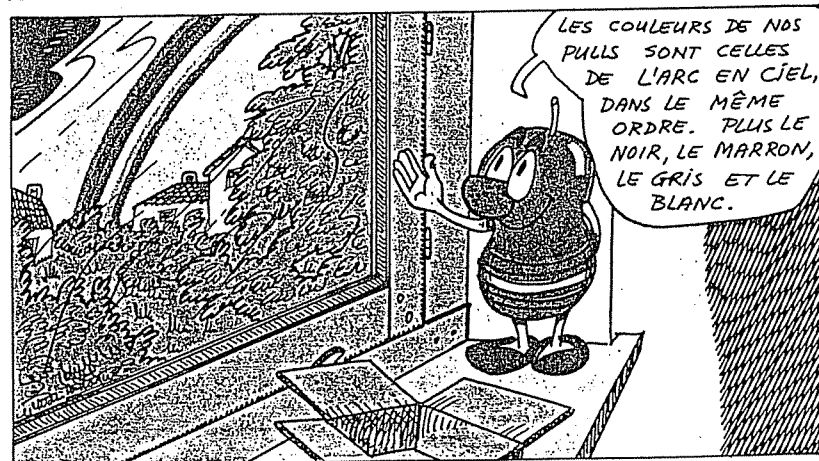
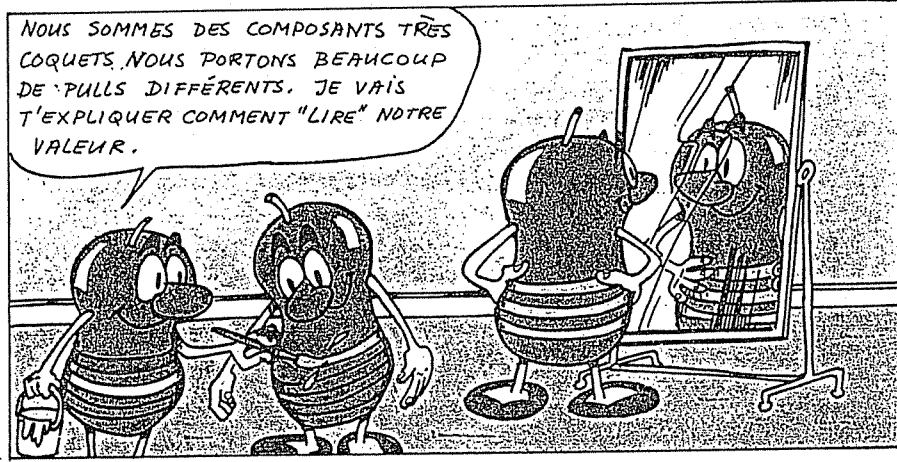
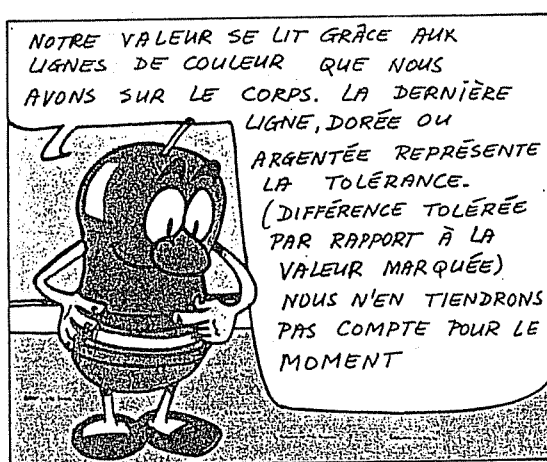
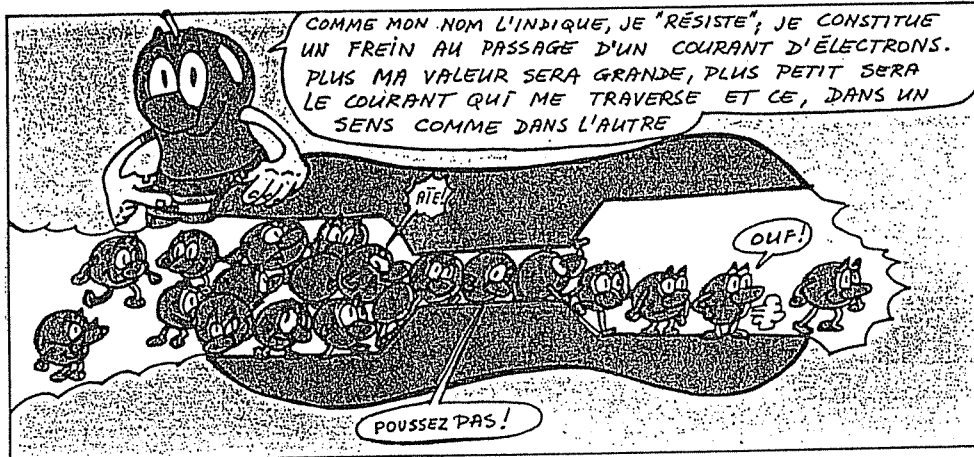
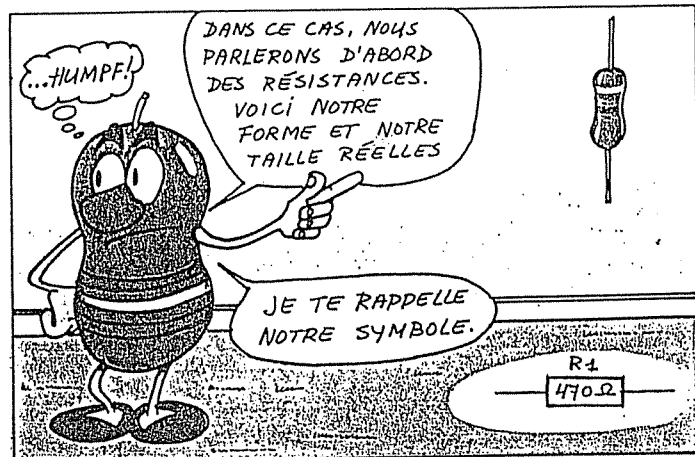
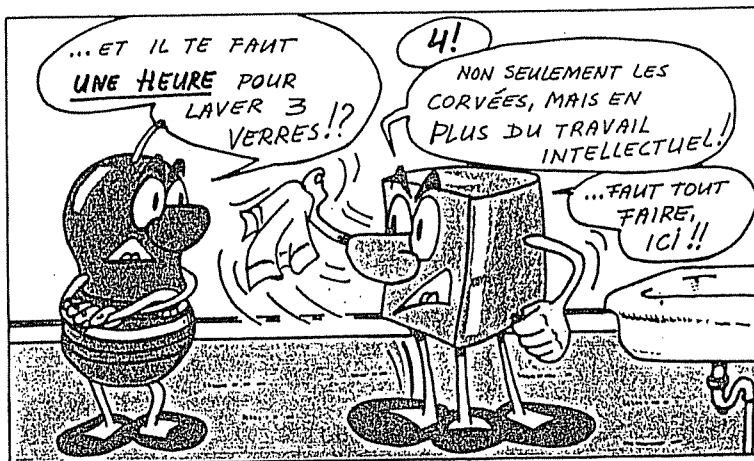
H_2SO_4 (LA MOLECULE D'ACIDE SULFURIQUE) EST FORMÉE DE 2 ATOMES D'HYDROGÈNE (H*) ET DE 4 ATOMES D'OXYGÈNE (O) COMBINÉS ENSEMBLE

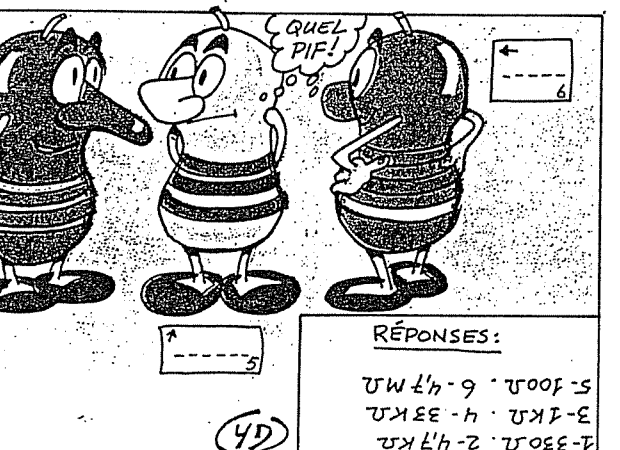
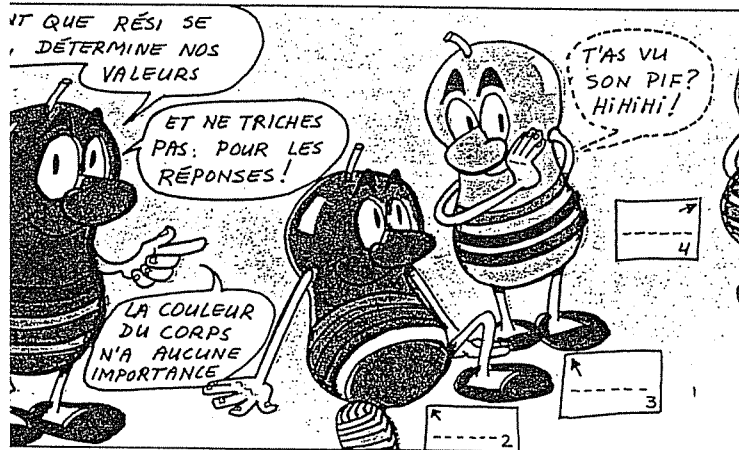
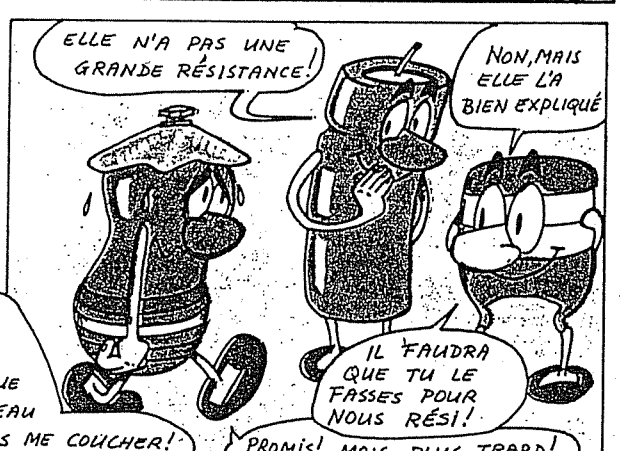
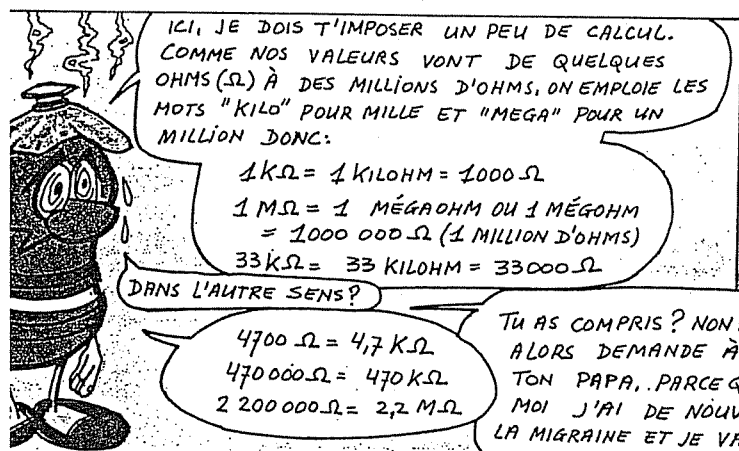
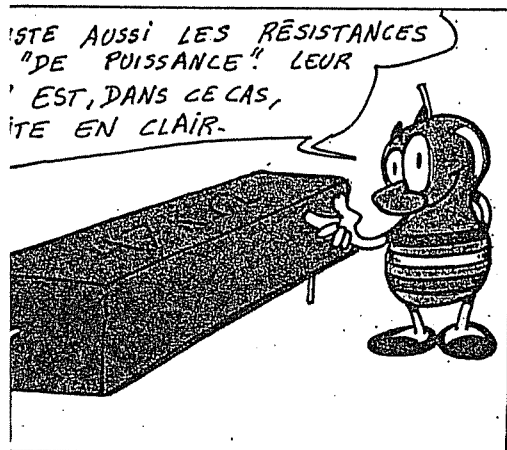
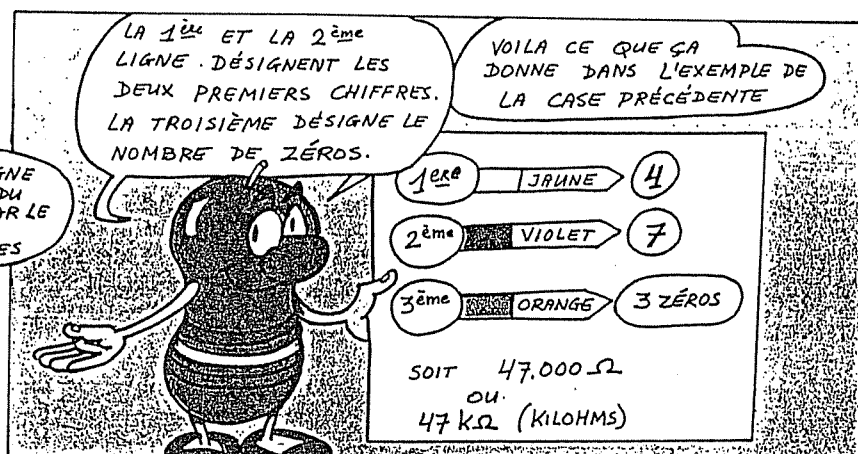
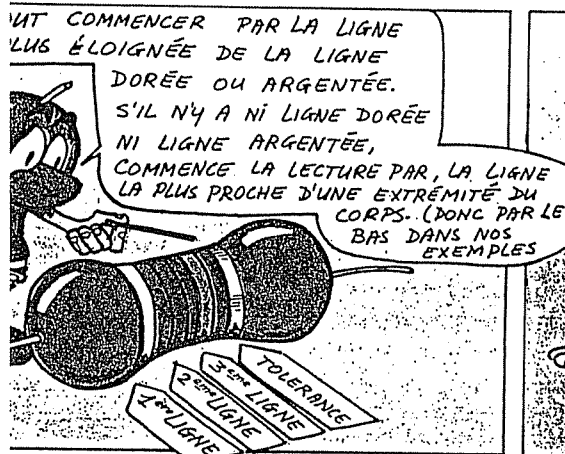
* H, S, O, CU, ZN, SONT DES SYMBOLES CHIMIQUES.

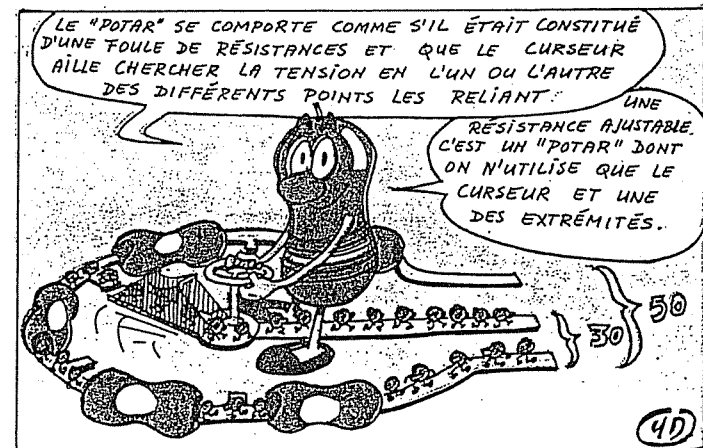
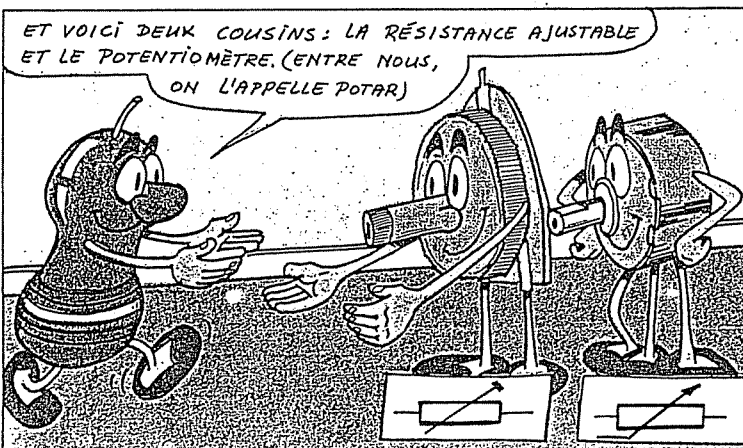
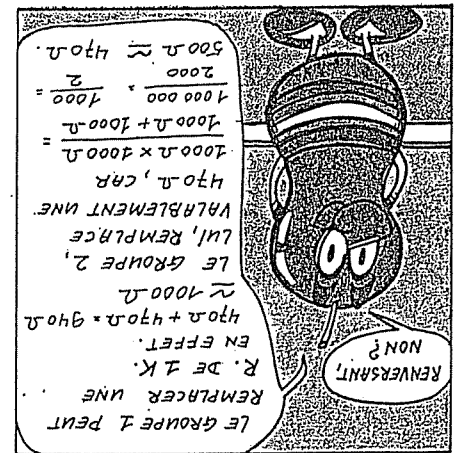
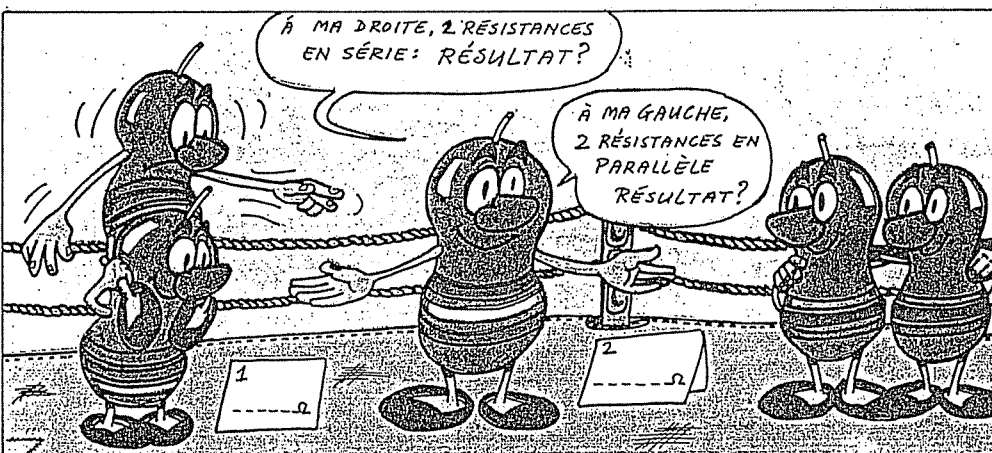
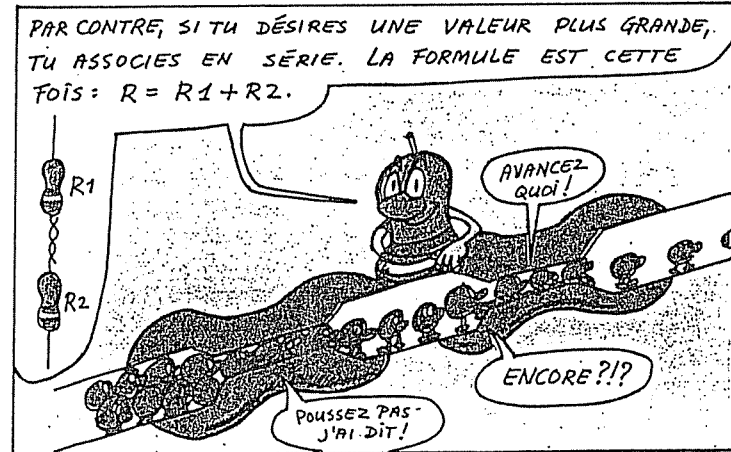
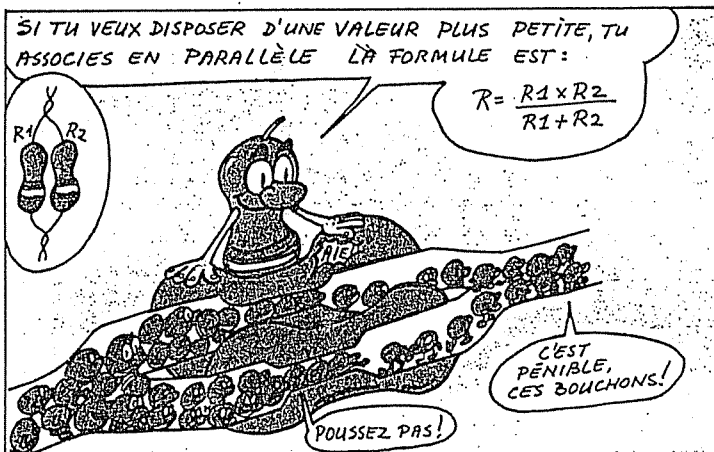
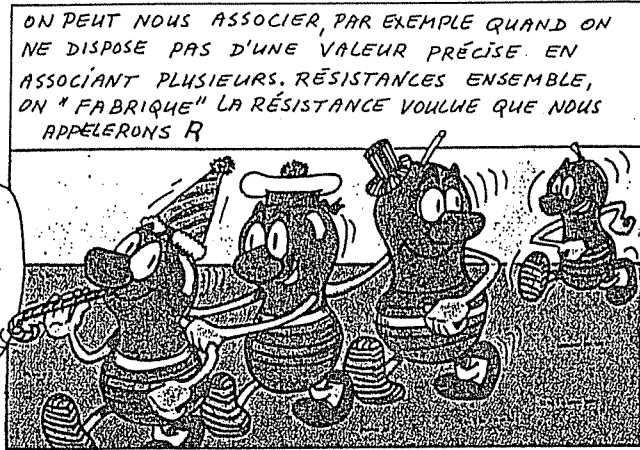
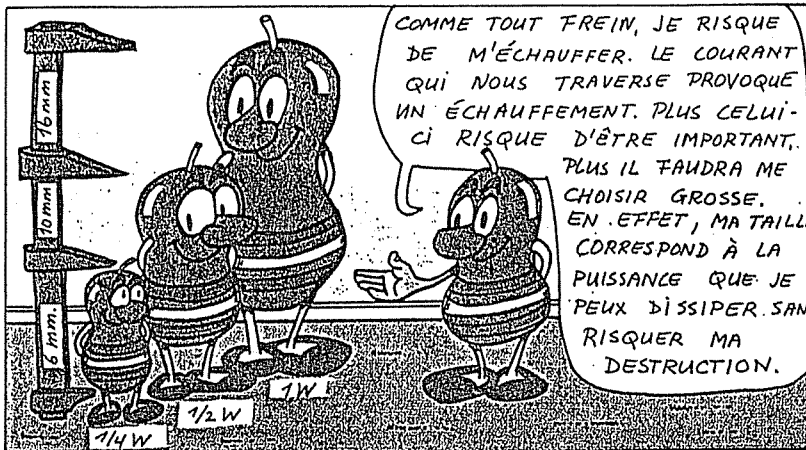


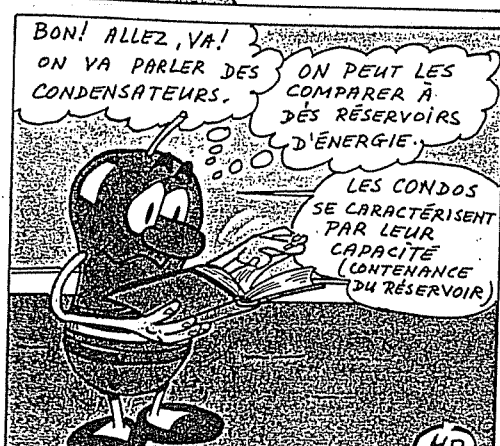
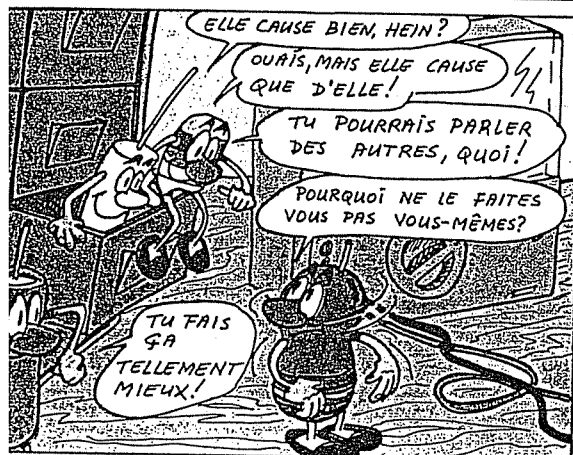
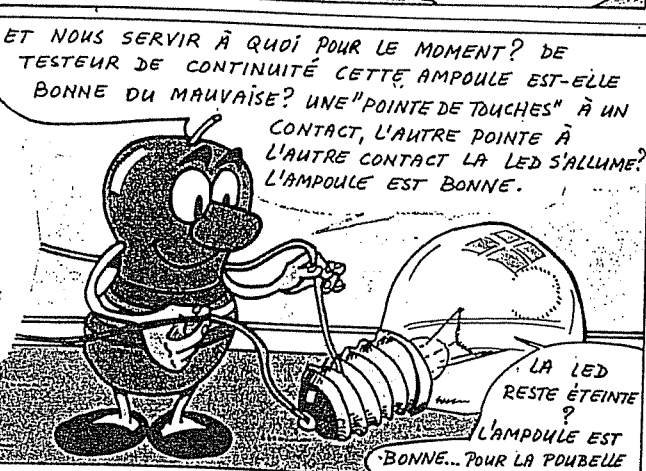
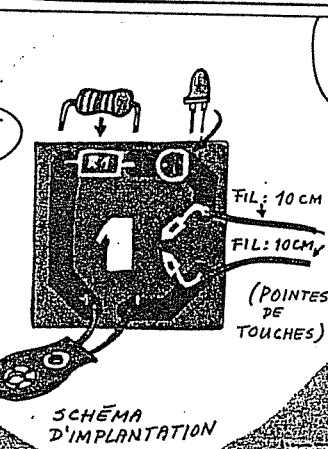
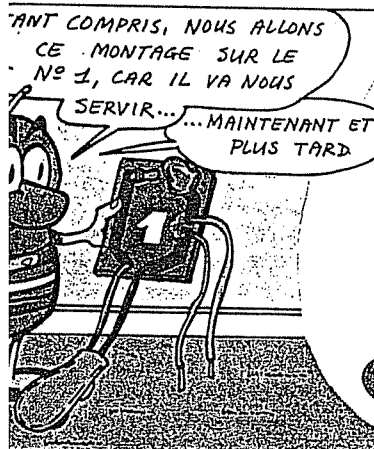
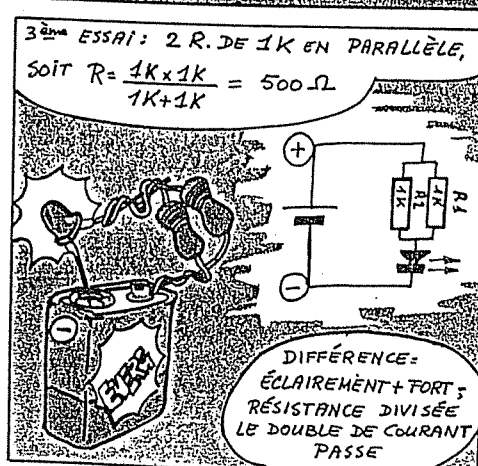
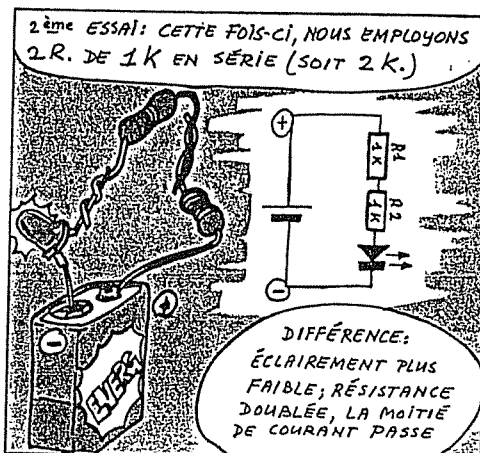
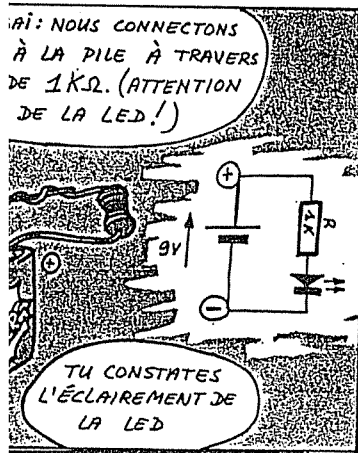
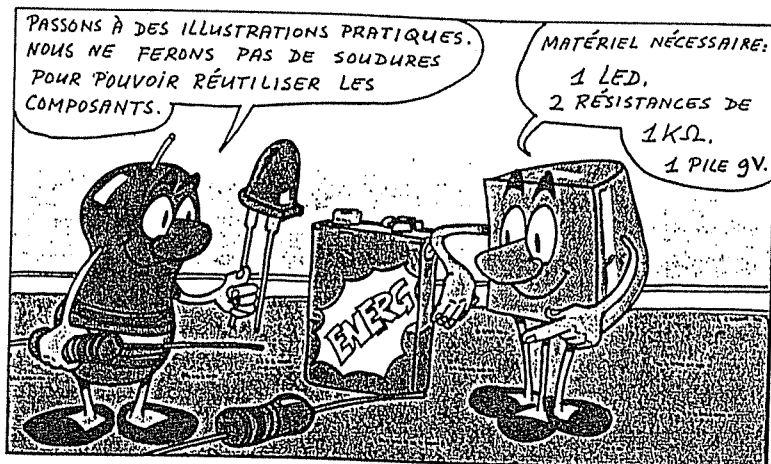
CHEZ ZN, CERTAINS ATOMES "S'ALLIENT" AVEC SO_4 (ABANDONNANT H) ET FORMENT $ZNSO_4$, SOIT DU SULFATE DE ZINC. CELUI-CI, PLUS LOURD, COULE AU FOND DU LIQUIDE.

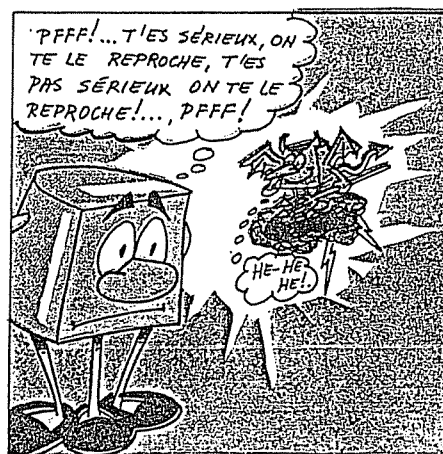
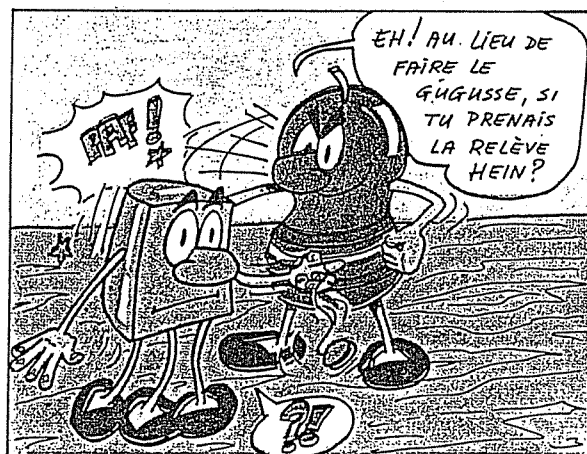
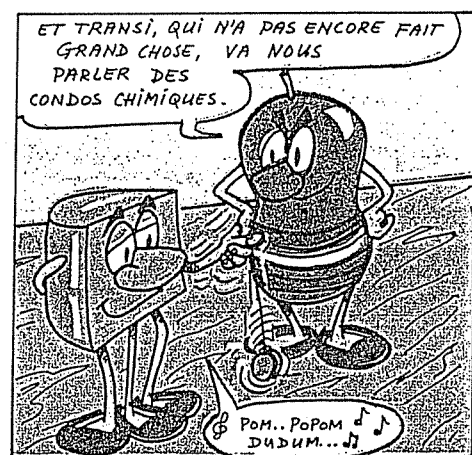
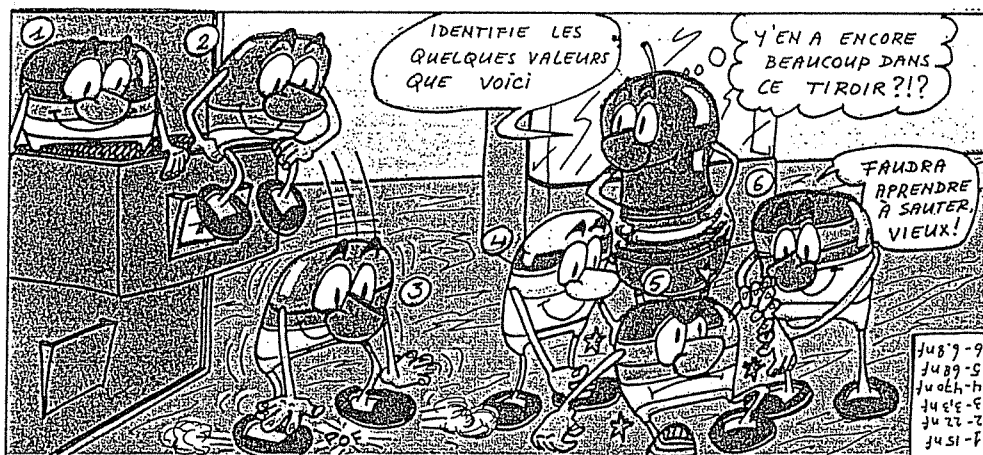
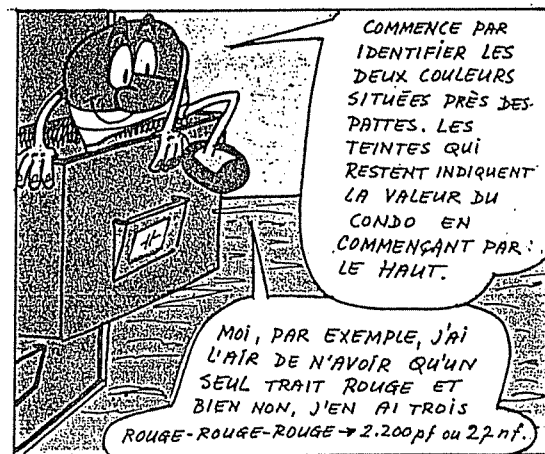
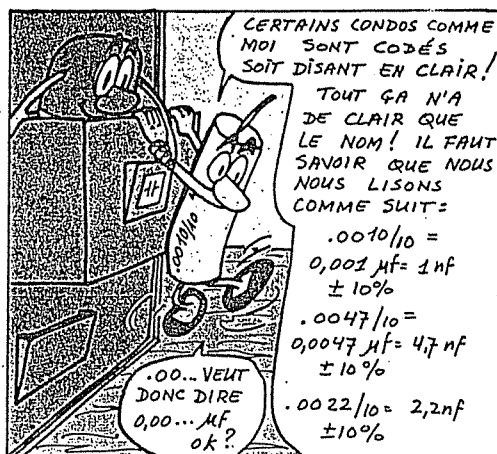
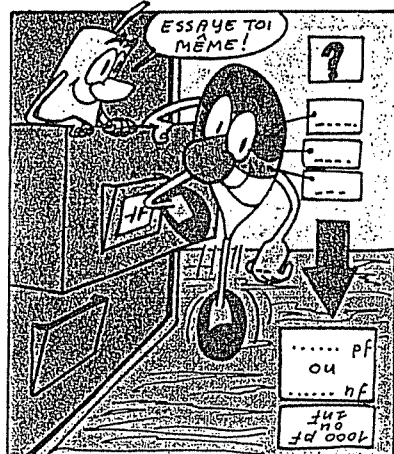
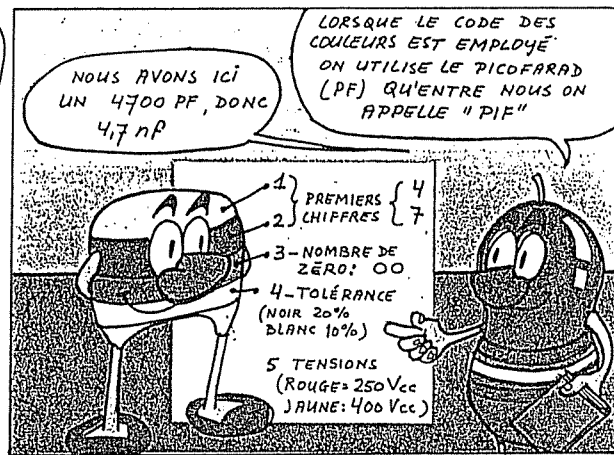
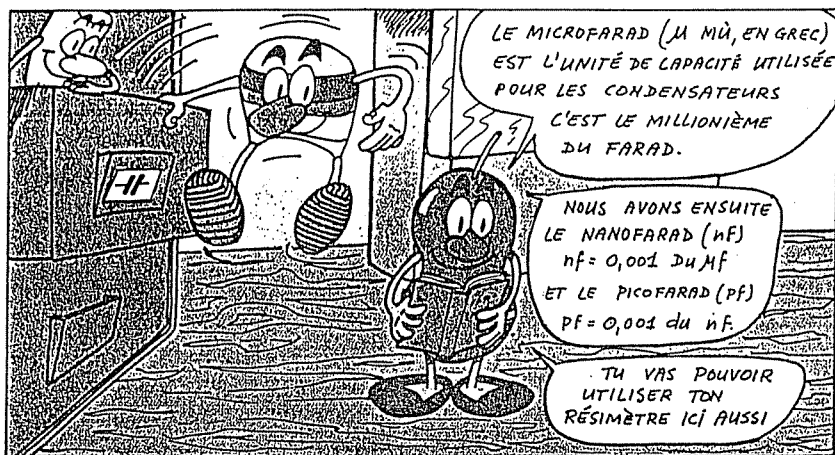


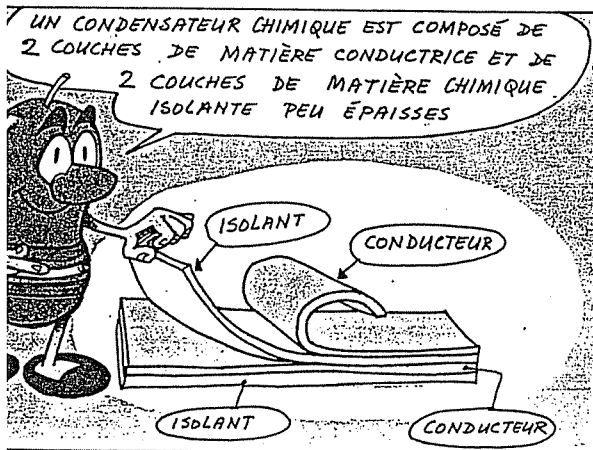
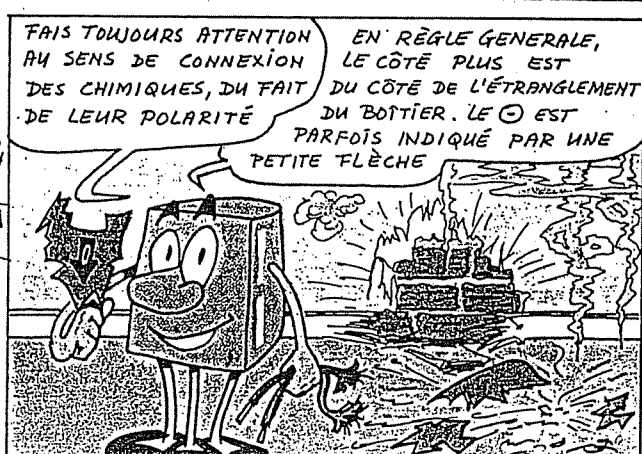
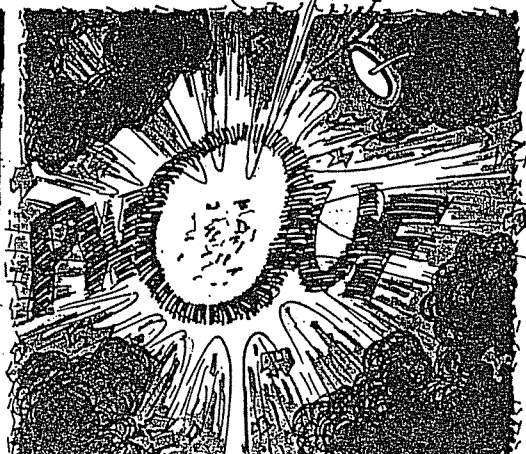
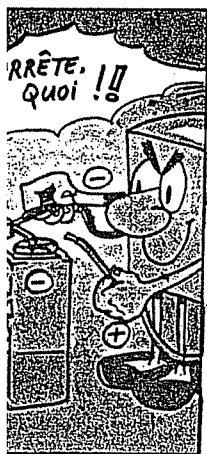
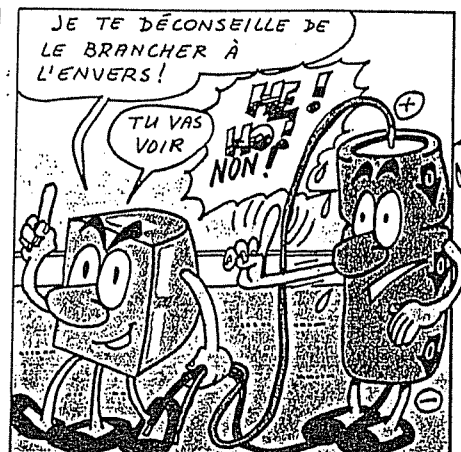


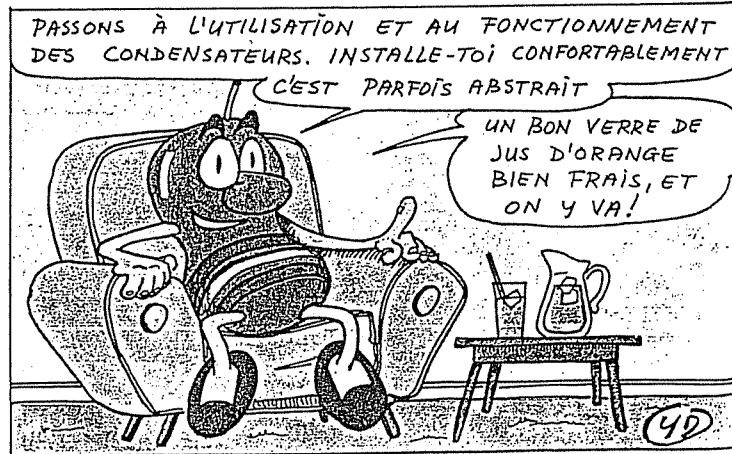
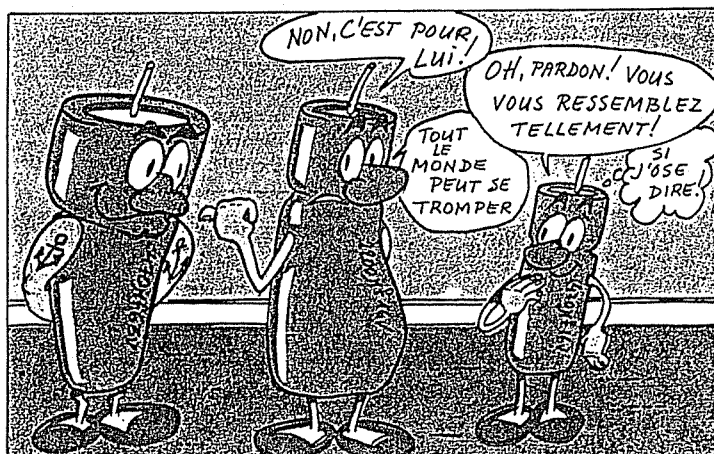
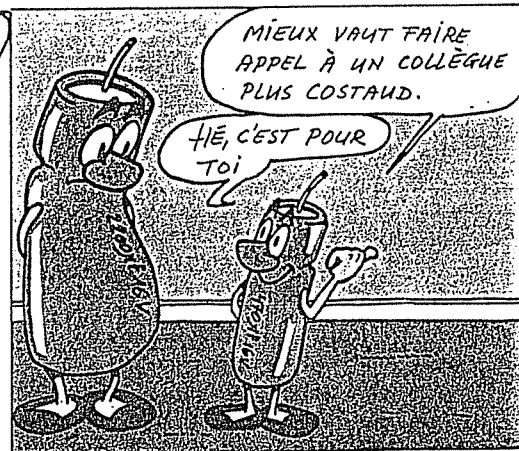
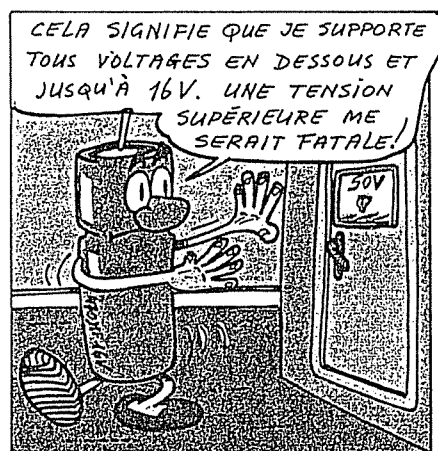
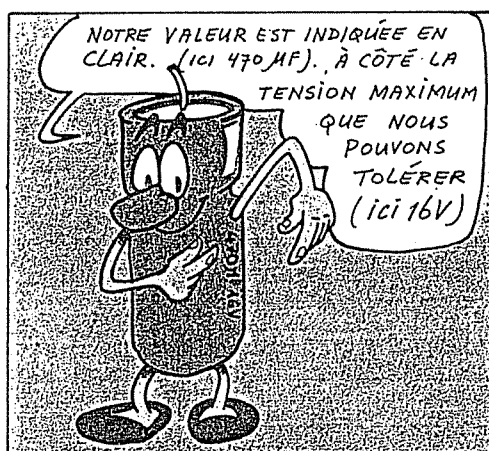
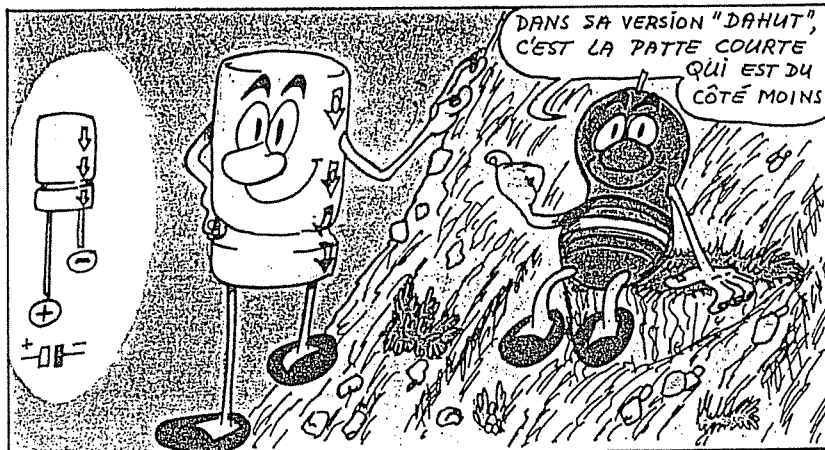
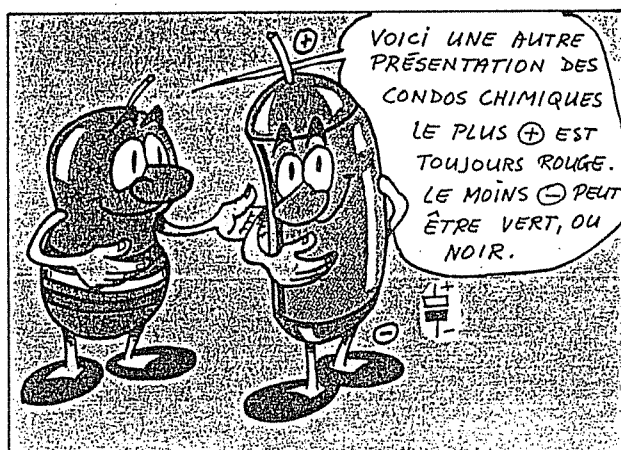
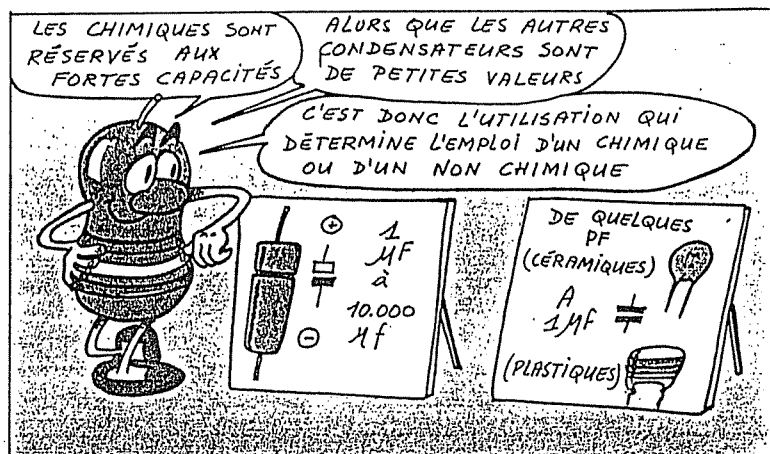
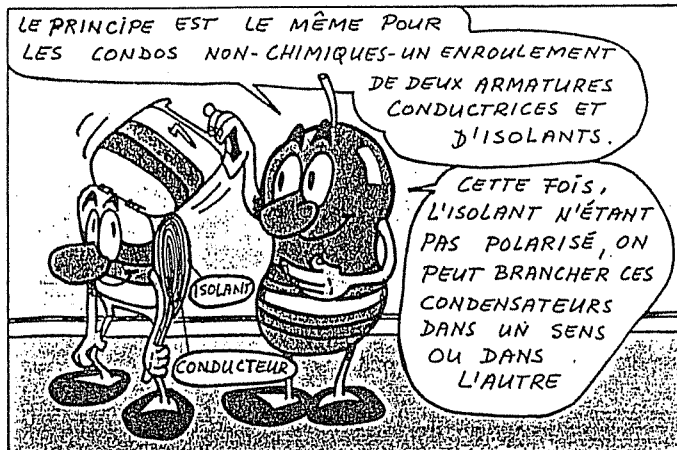


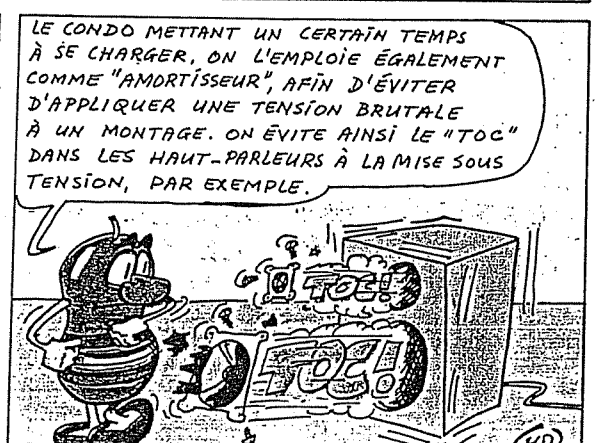
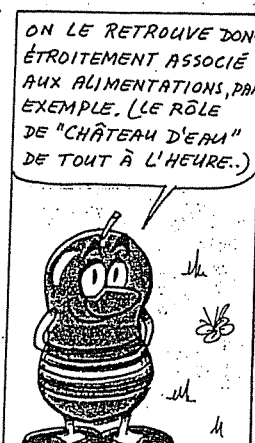
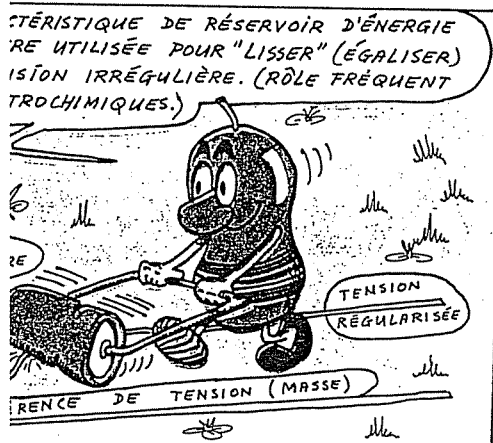
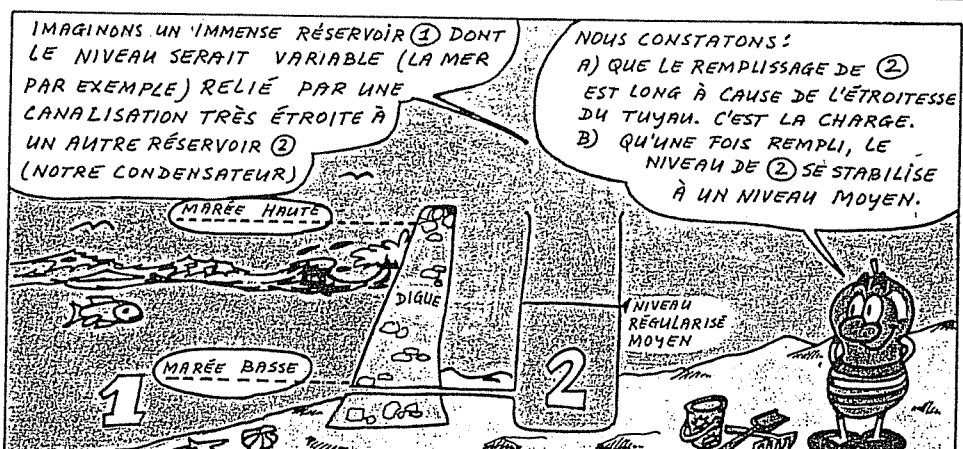
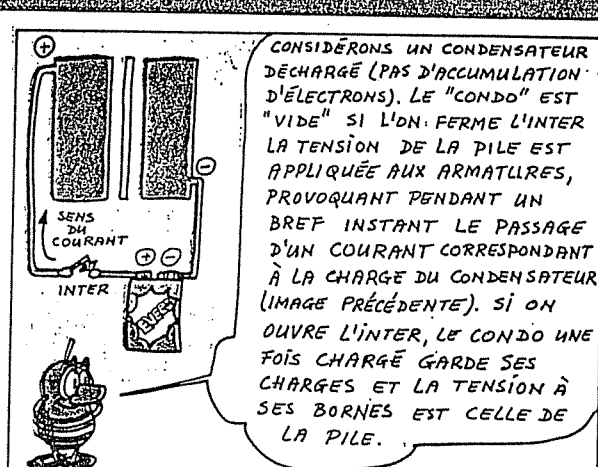
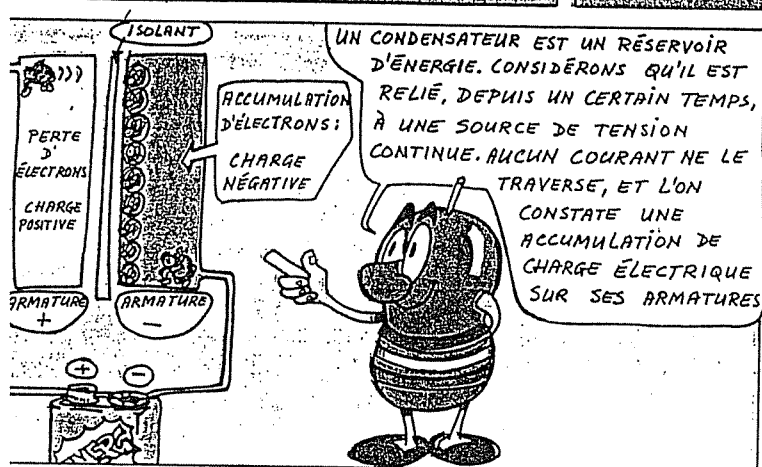
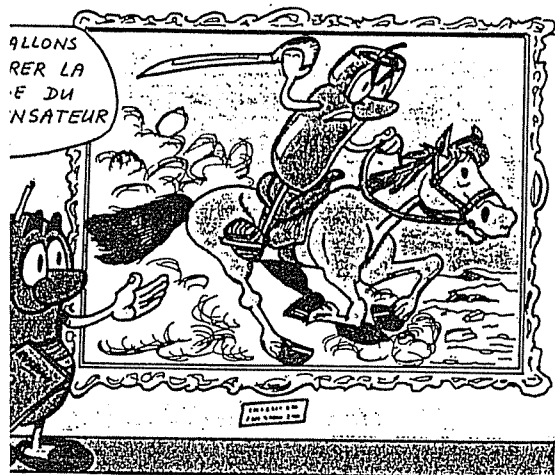








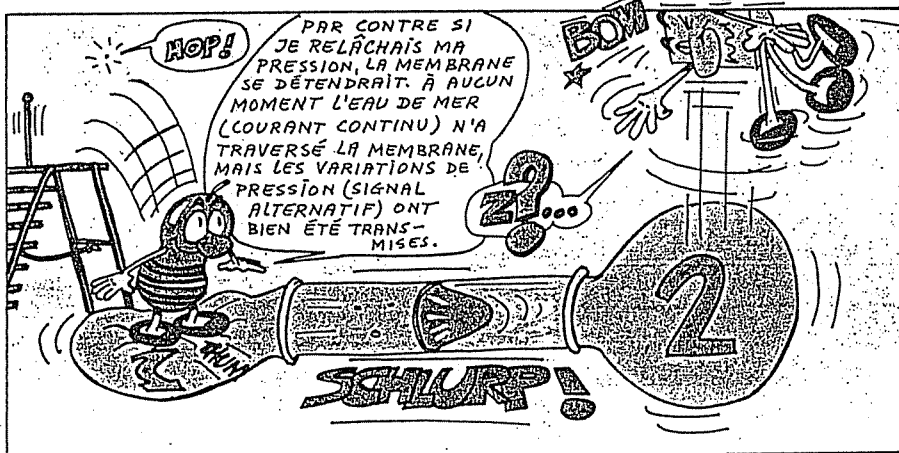
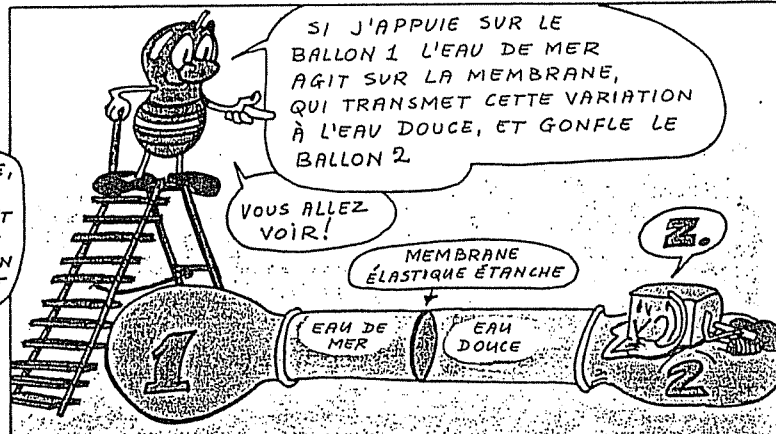




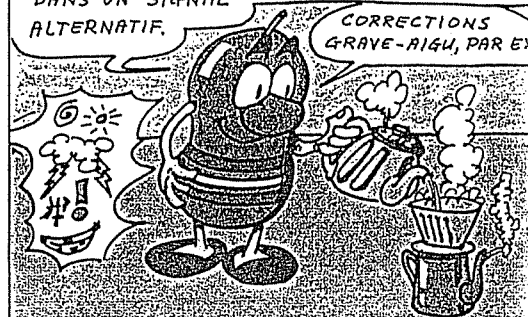
ON L'EMPLOIE AUSSI COMME ÉLÉMENT DE LIASON. IL TRANSMET DES VARIATIONS DE TENSION (SIGNAL ALTERNATIF) TOUT EN BLOQUANT LE CONTINU. TOUTE FOIS, SA VALEUR DOIT ÊTRE CALCULÉE EN FONCTION DE LA FRÉQUENCE À TRANSMETTRE.

TU TE SOUVIENS DE LA FRÉQUENCE? ON EN A PARLÉ AU CHAPITRE ÉLECTRICITÉ: ELLE S'EXPRIME EN HERTZ (HZ), C'EST LE NOMBRE DE CYCLES DURANT UNE SECONDE.

MAIS JE PARLE, JE PARLE.... VOYONS PLUTÔT À CÔTÉ UNE ILLUSTRATION DE TOUT ÇA...

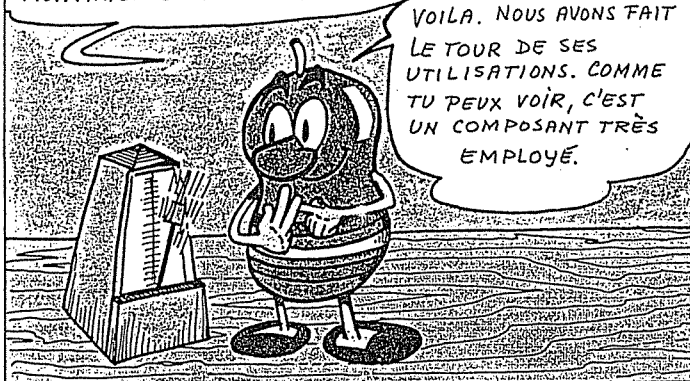


LE CONDO EST TRÈS UTILISÉ DANS CES MONTAGES APPELÉS "FILTRES" DONT LE RÔLE EST DE SÉLECTIONNER CERTAINES FRÉQUENCES PARMI D'AUTRES CONTENUES DANS UN SIGNAL ALTERNATIF.



IL EST ENFIN EMPLOYÉ DANS LES MONTAGES OSCILLATEURS.

VOILA. NOUS AVONS FAIT LE TOUR DE SES UTILISATIONS. COMME TU PEUX VOIR, C'EST UN COMPOSANT TRÈS EMPLOYÉ.

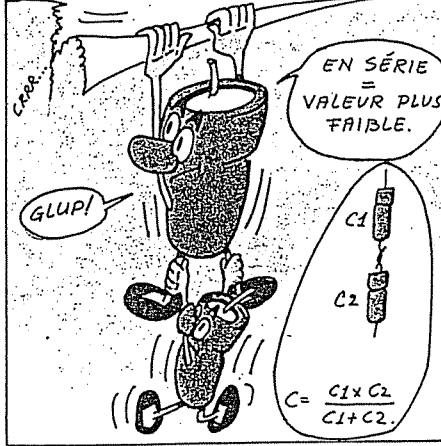
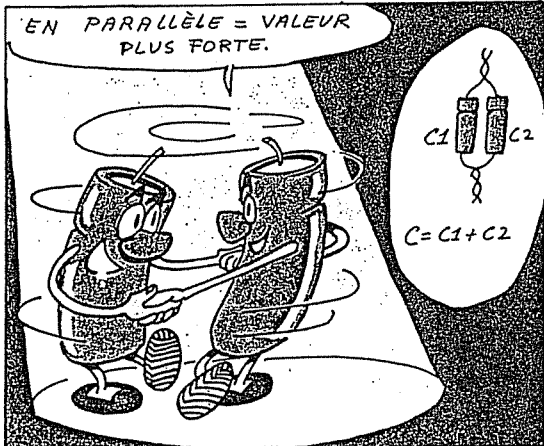


LES CONDENSATEURS PEUVENT, COMME LES RÉISTANCES, ÊTRE ASSOCIÉS ENTRE EUX, MAIS LES FORMULES DE CALCUL SONT L'INVERSE DE CELLES DES RÉISTANCES.



ON ASSOCIE LES CONDENSATEURS POUR LES MÊMES RAISONS QUE LES RÉISTANCES.

EN PARALLÈLE = VALEUR PLUS FORTE.



UN PETIT EXERCICE:

A EN SÉRIE AVEC B ÉGAL?

$C = \frac{C_1 \times C_2}{C_1 + C_2}$

